

Printer Dover

Spruce version 11.2 -- spooler version 11.2

File: MemD-Rev-Ca.sil etc.

Creation date: 30-Oct-79 14:11:21

For: Spruce

25 total sheets = 24 pages, 1 copy.

Problems encountered:

Font HELVETICA24B substituted for font HELVE24.

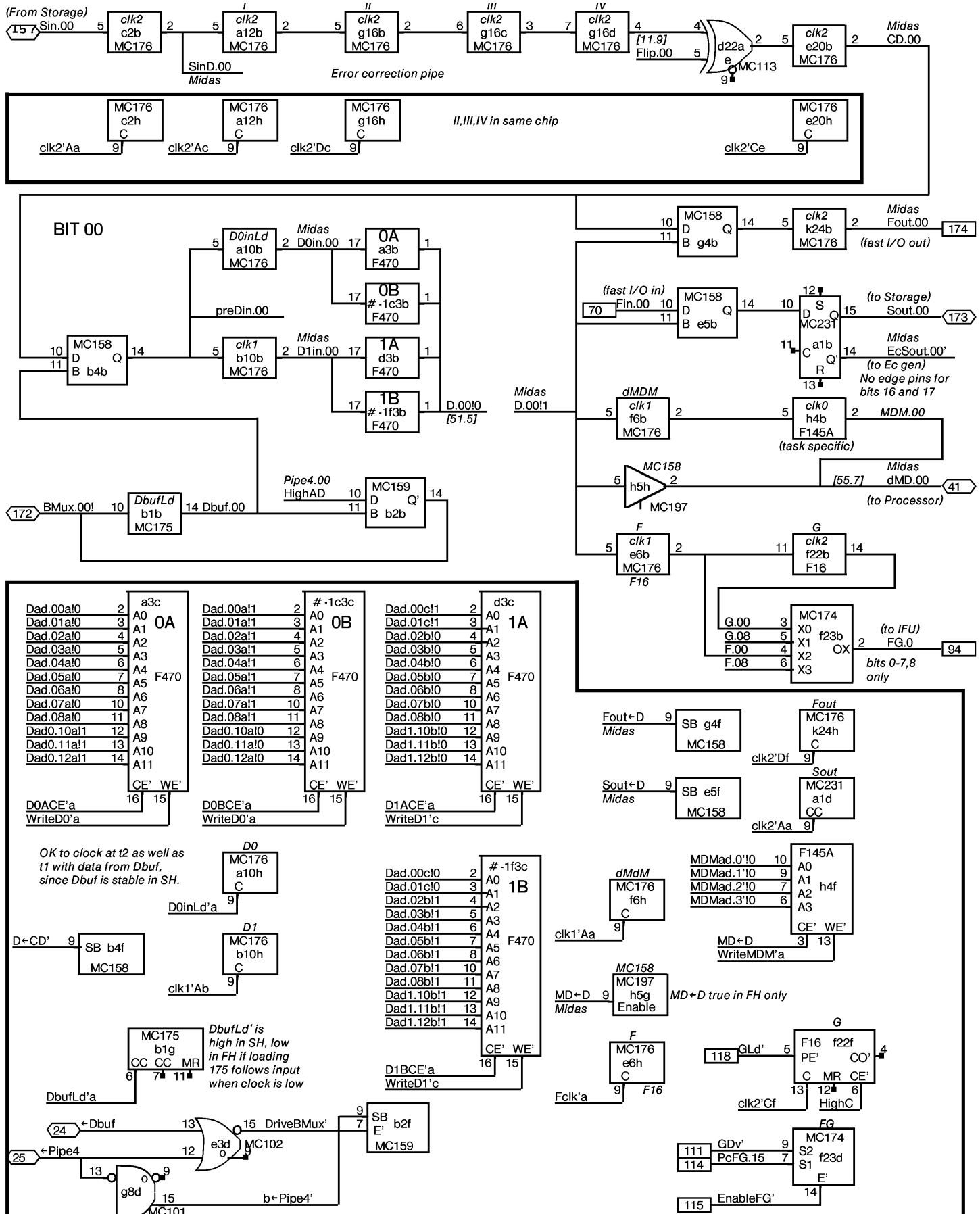
DORADO SCHEMATICS

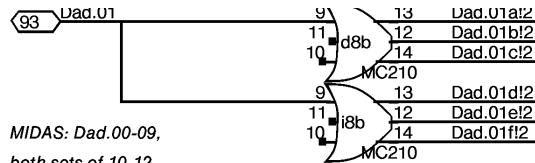
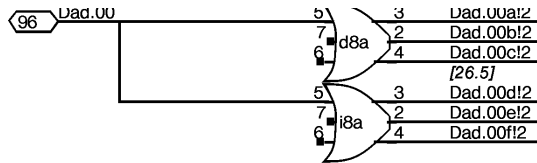
Memory Data Board

(4K chip version)

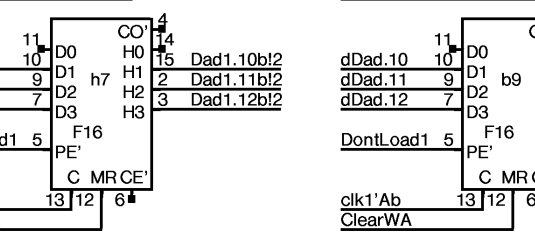
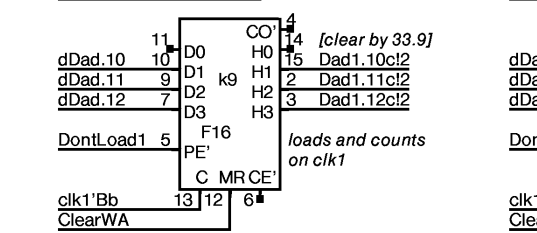
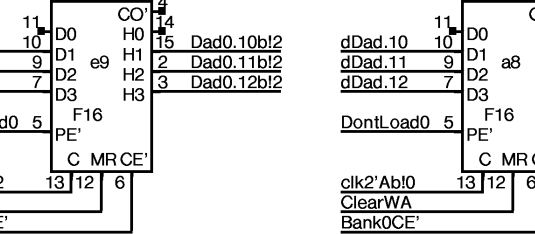
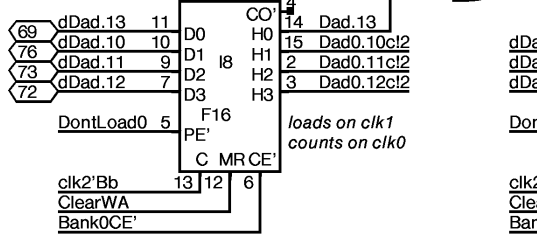
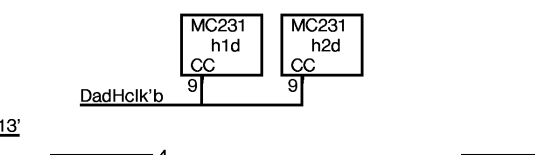
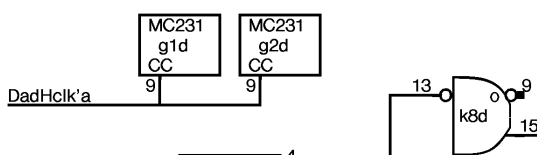
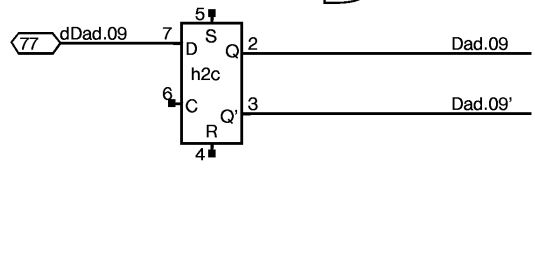
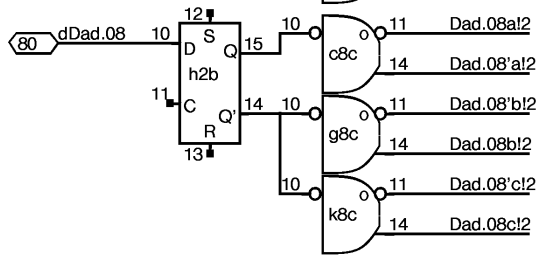
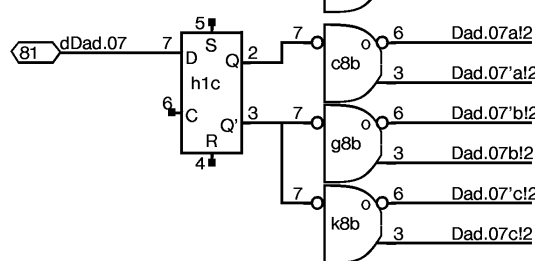
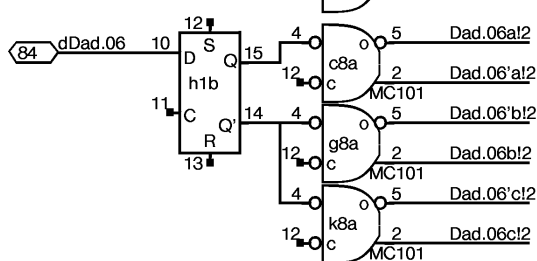
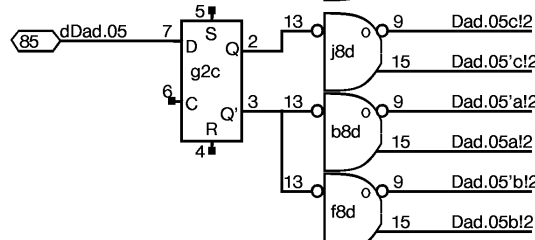
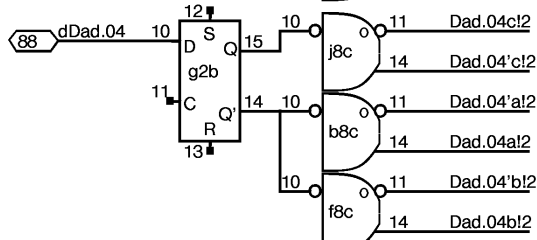
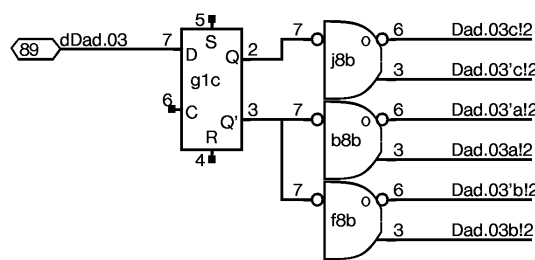
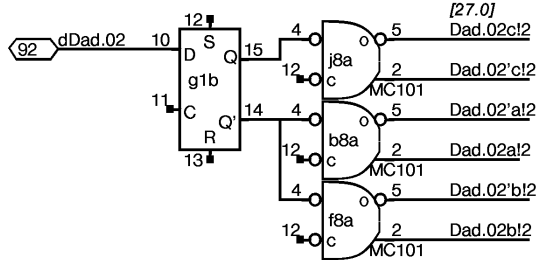
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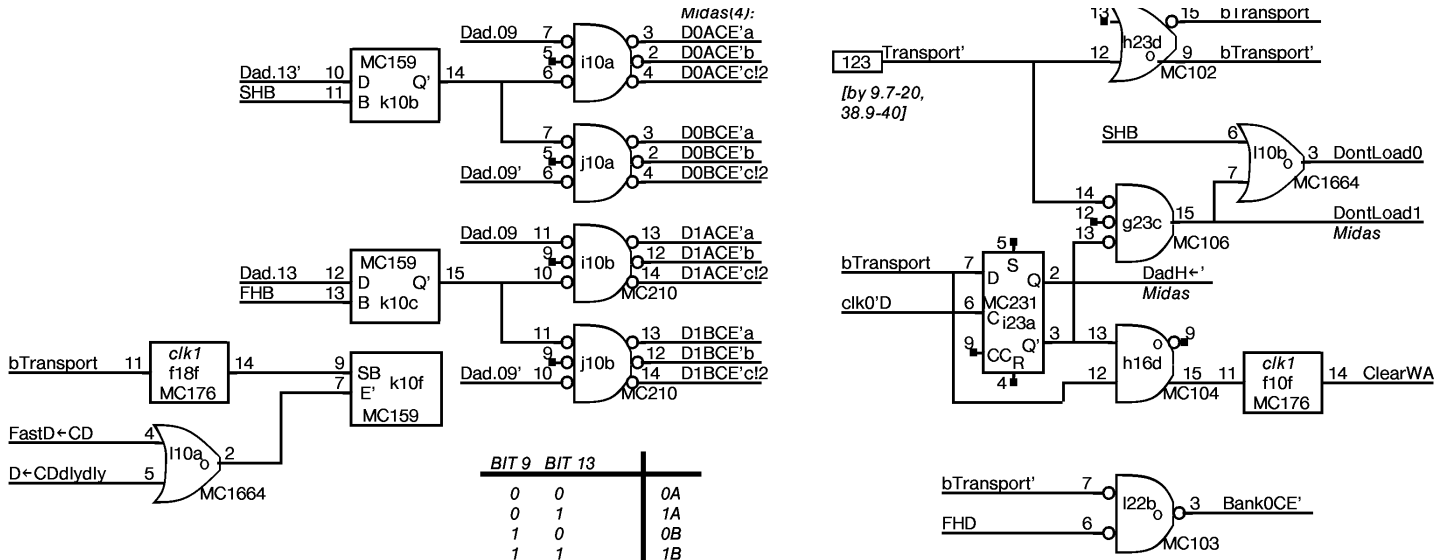
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Control, D memory CE's and Write's _____	03
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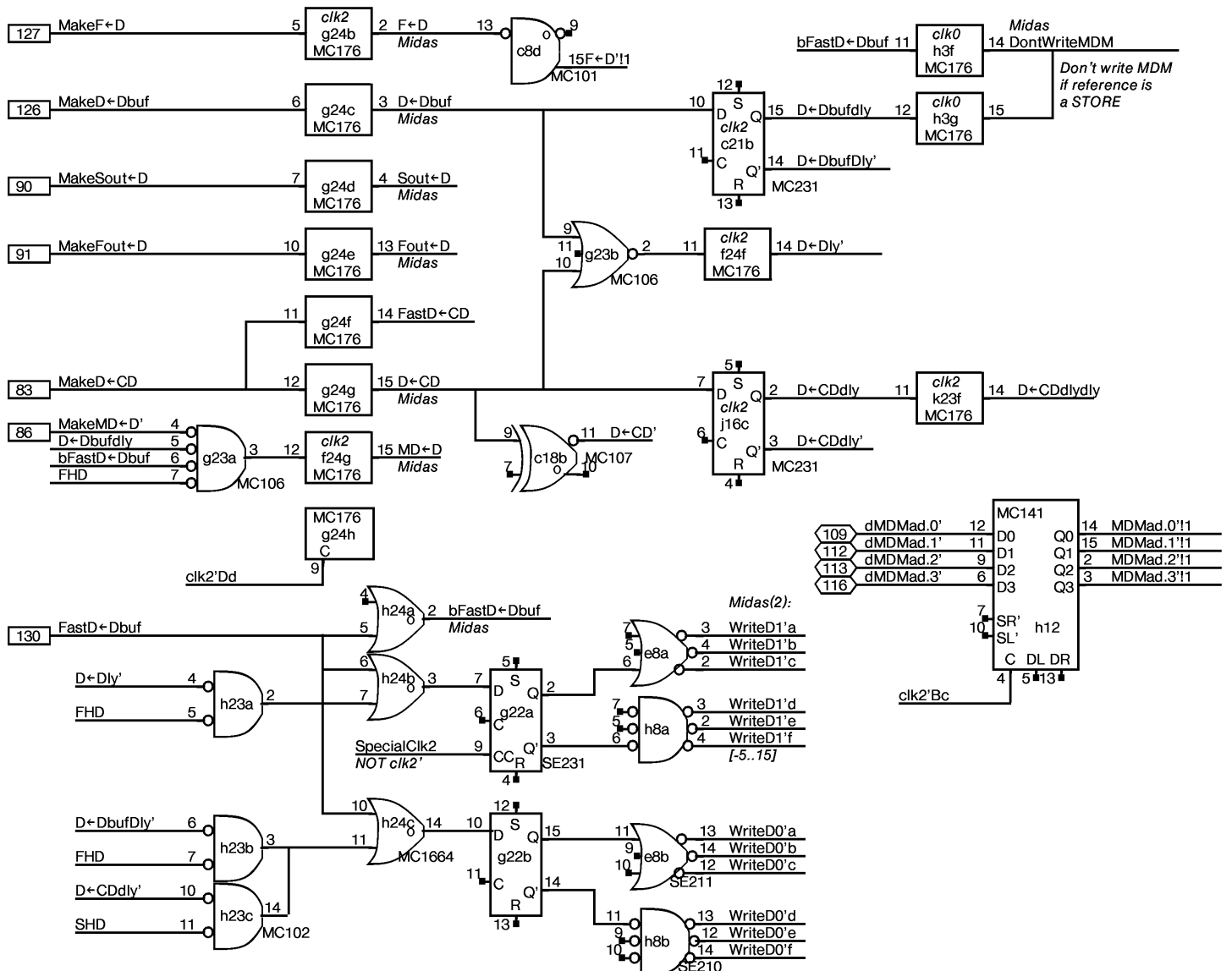


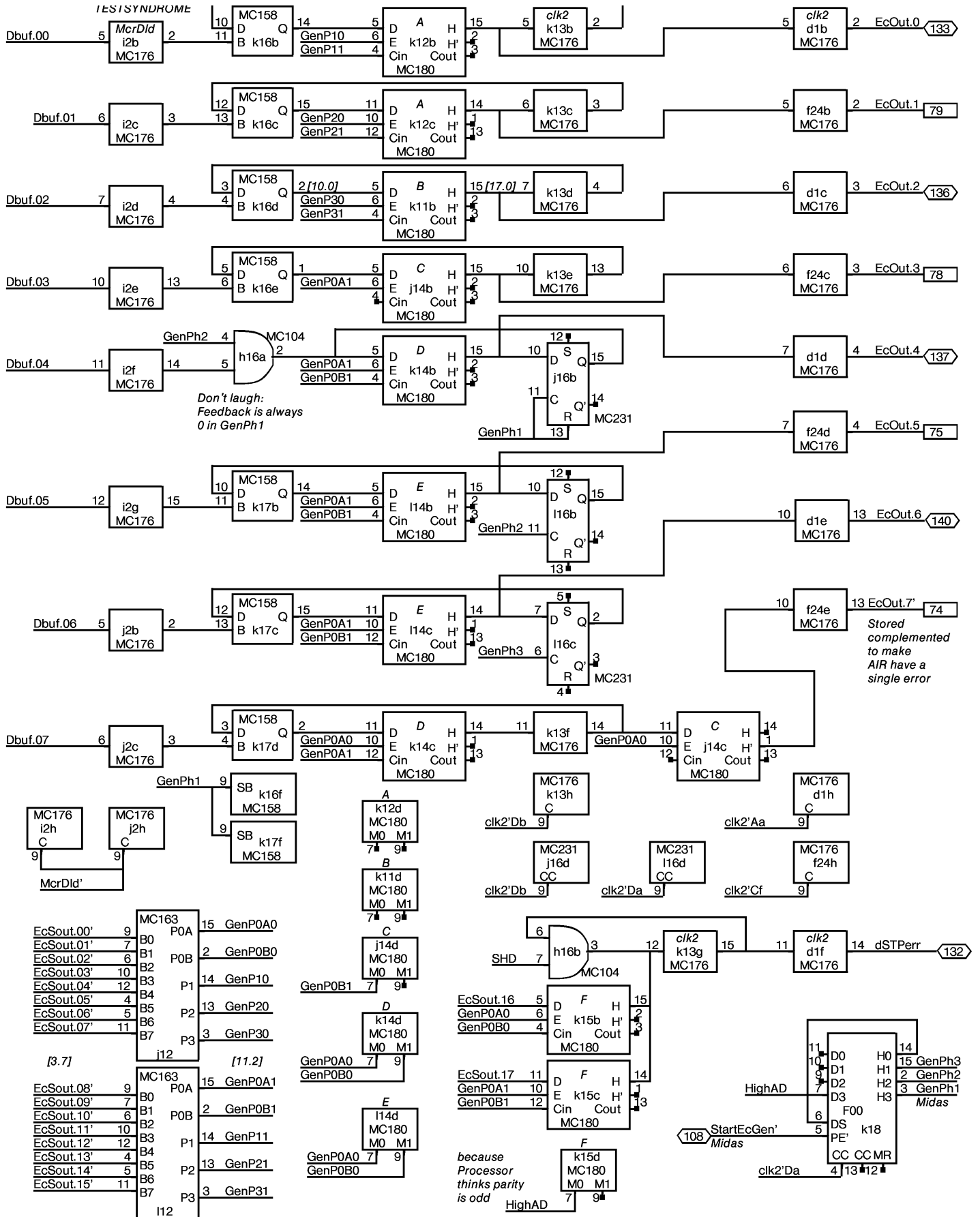
MIDAS: Dad.00-09,
both sets of 10-12

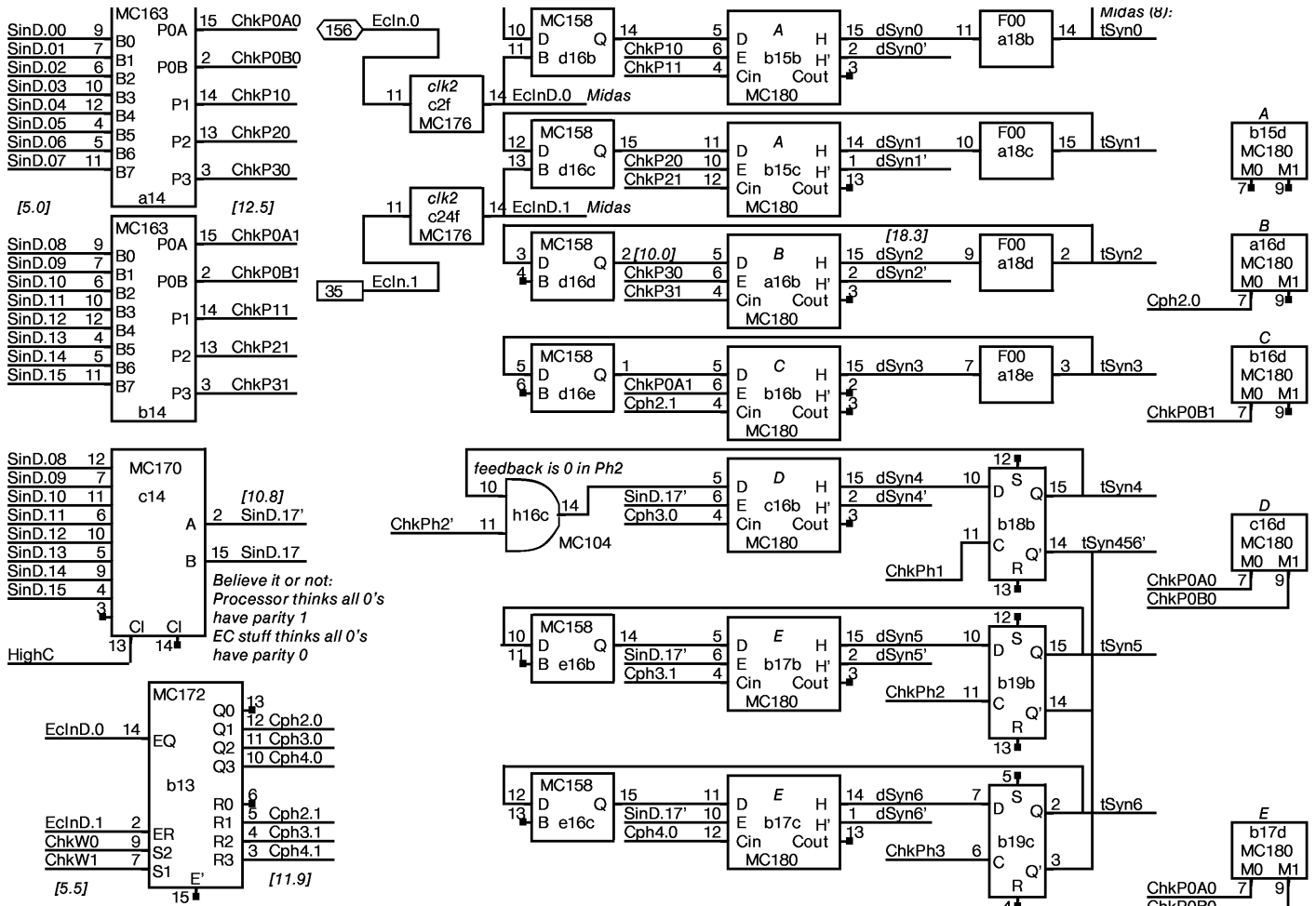




Midas
MakeMDM+D'
87 qualifies clock directly







Note: the MC180 is a dual 5-input xor gate, in which the M0 and M1 inputs are common.

Stored Ci are retrieved two at a time, and xored into the syndrome as they arrive. The Cphi.0/1 signals are the arriving Ci in the proper phase, 0 otherwise

MC163 operation

	0	1	2	3	4	5	6	7
P1	x	x	x	x	x	x	x	x
P2	x	x	x	x	x	x	x	x
P3	x	x	x	x	x	x	x	x
P0A	x	x	x	x	x	x	x	x
P0B	x	x	x	x	x	x	x	x

EC code being used:

	0	1	2	3	4	5	6	7
C0	x	x	x	x	x	x	x	x
C1	x	x	x	x	x	x	x	x
C2	x	x	x	x	x	x	x	x
C3	x	x	x	x	x	x	x	x
C4	x	x	x	x	x	x	x	x
C5	x	x	x	x	x	x	x	x
C6	x	x	x	x	x	x	x	x
C7	x	x	x	x	x	x	x	x

C0-2 the same in each byte

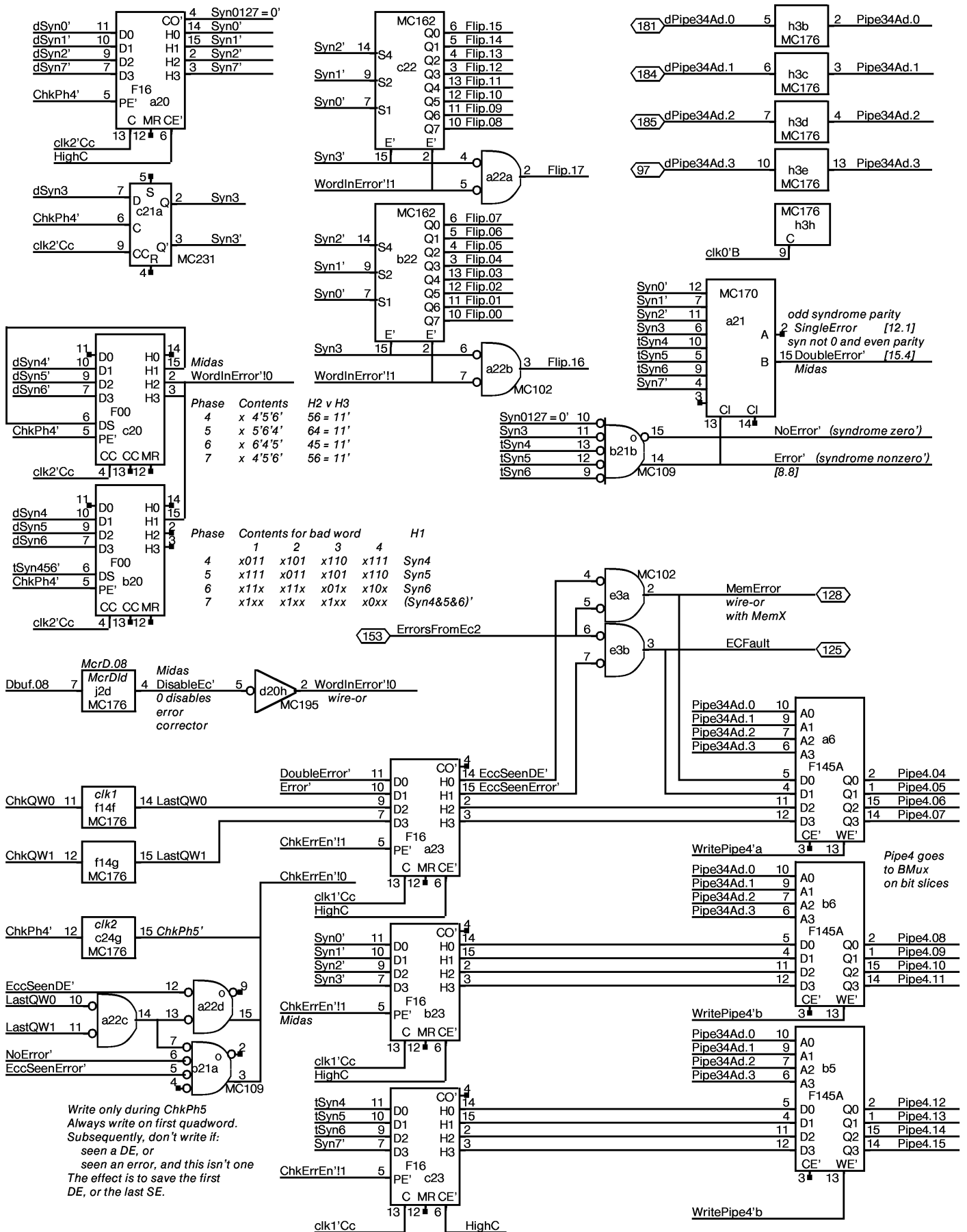
	0	8	16	24	32	40	48	56
C3	x	x	x	x	x	x	x	x
C4	x	x	x	x	x	x	x	x
C5	x	x	x	x	x	x	x	x
C6	x	x	x	x	x	x	x	x
C7	+	-	+	-	+	-	+	-

(+ repeats 0-7
- complements it)

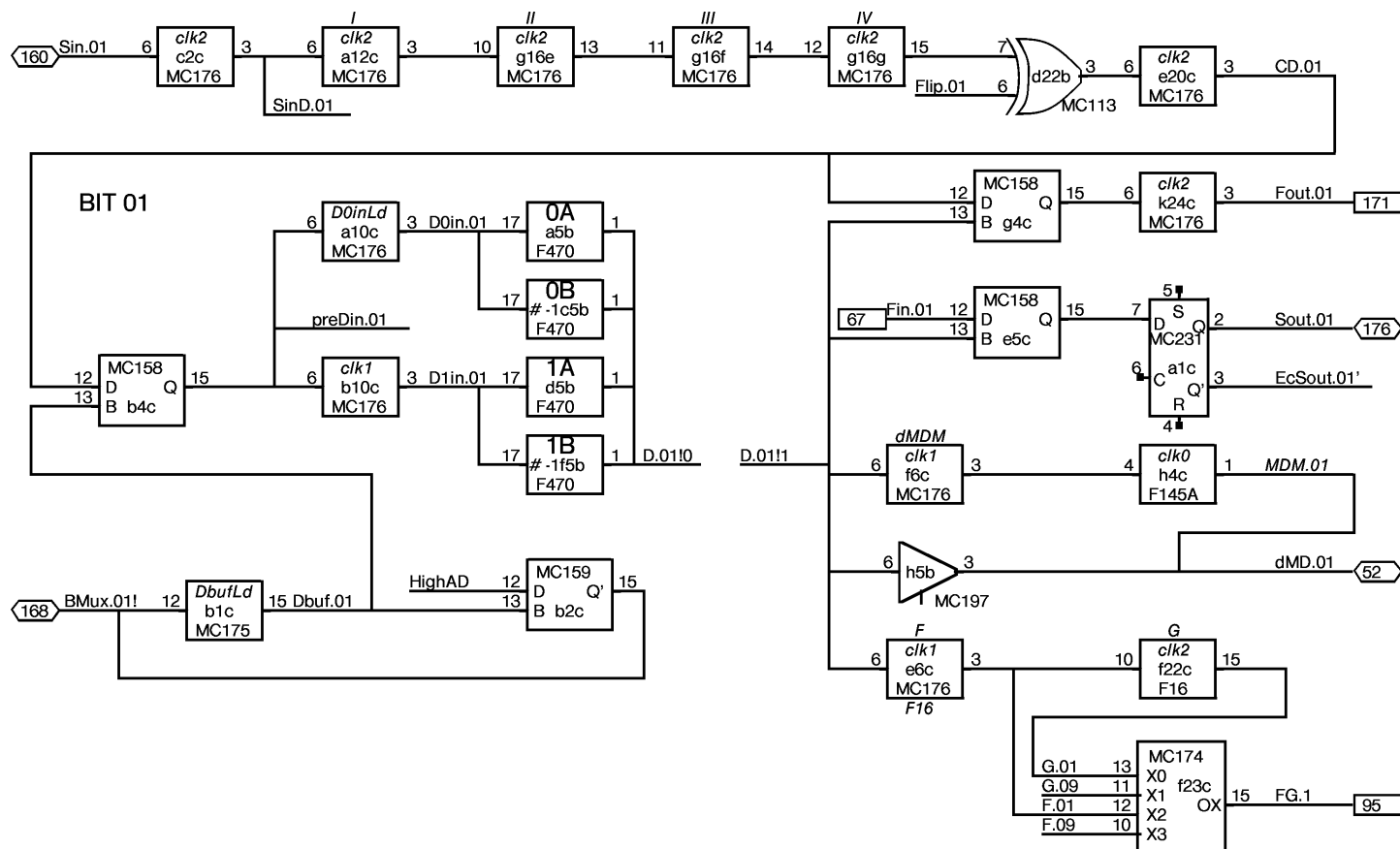
Syndrome is xor of retrieved Ci and Ci computed on retrieved data. Hence it is the Ci for the error pattern, and the following facts follow from looking at the pictures, which show the Ci for single-bit patterns. Syn0-3 point to erroneous bit. Syn4-6 point to bad word with a 2-out-of-3 code:

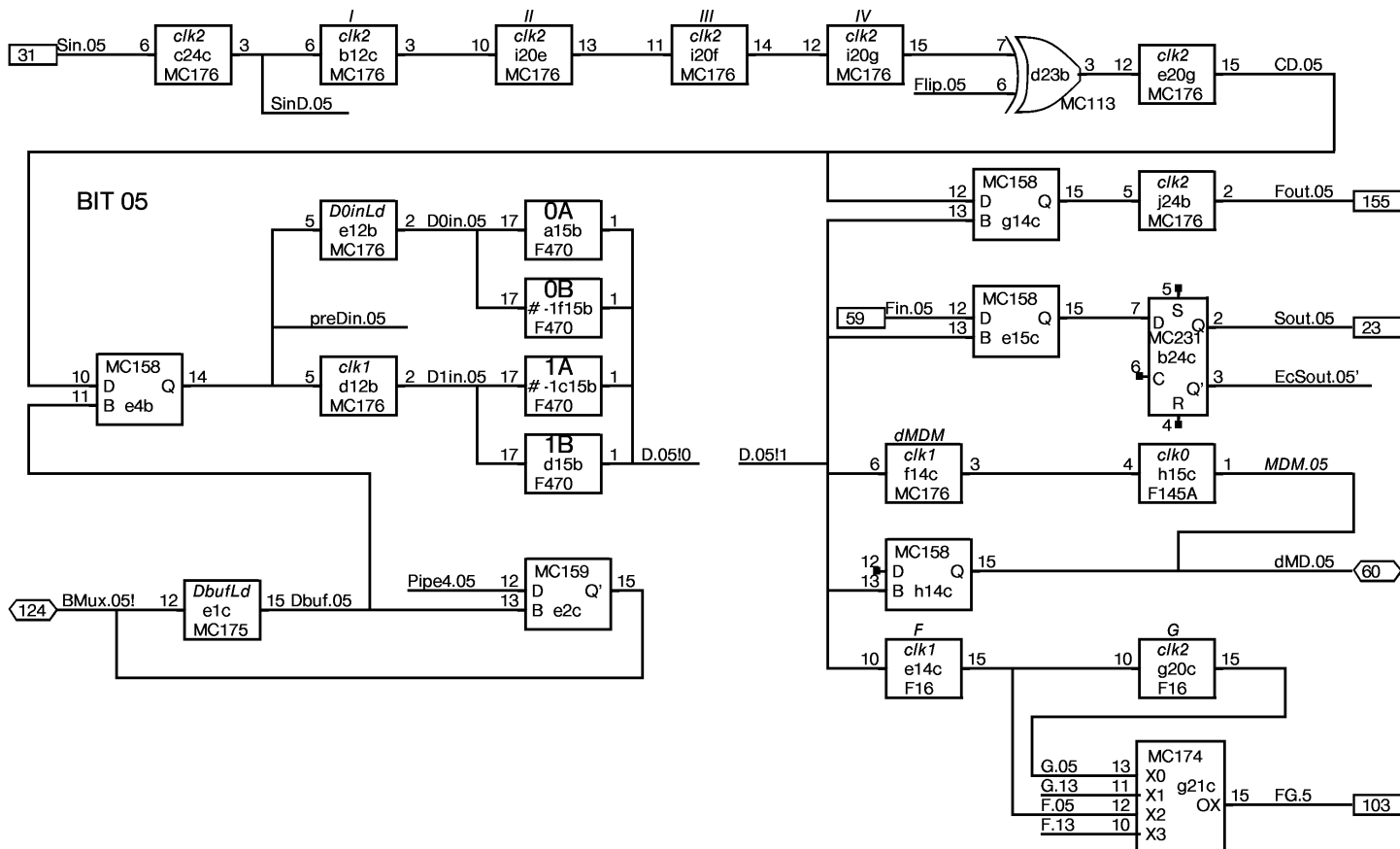
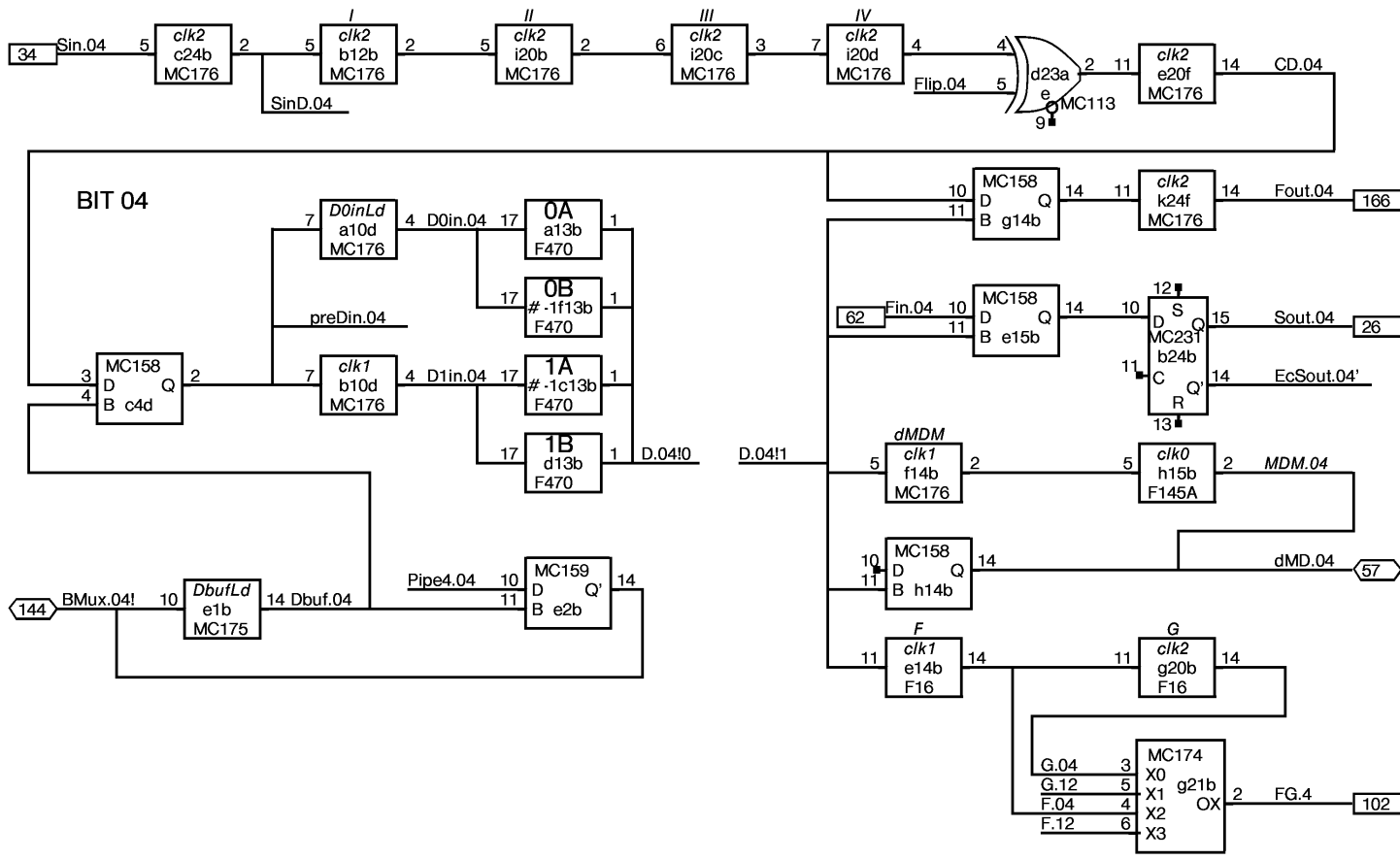
word 1 011
word 2 101
word 3 110
word 4 111

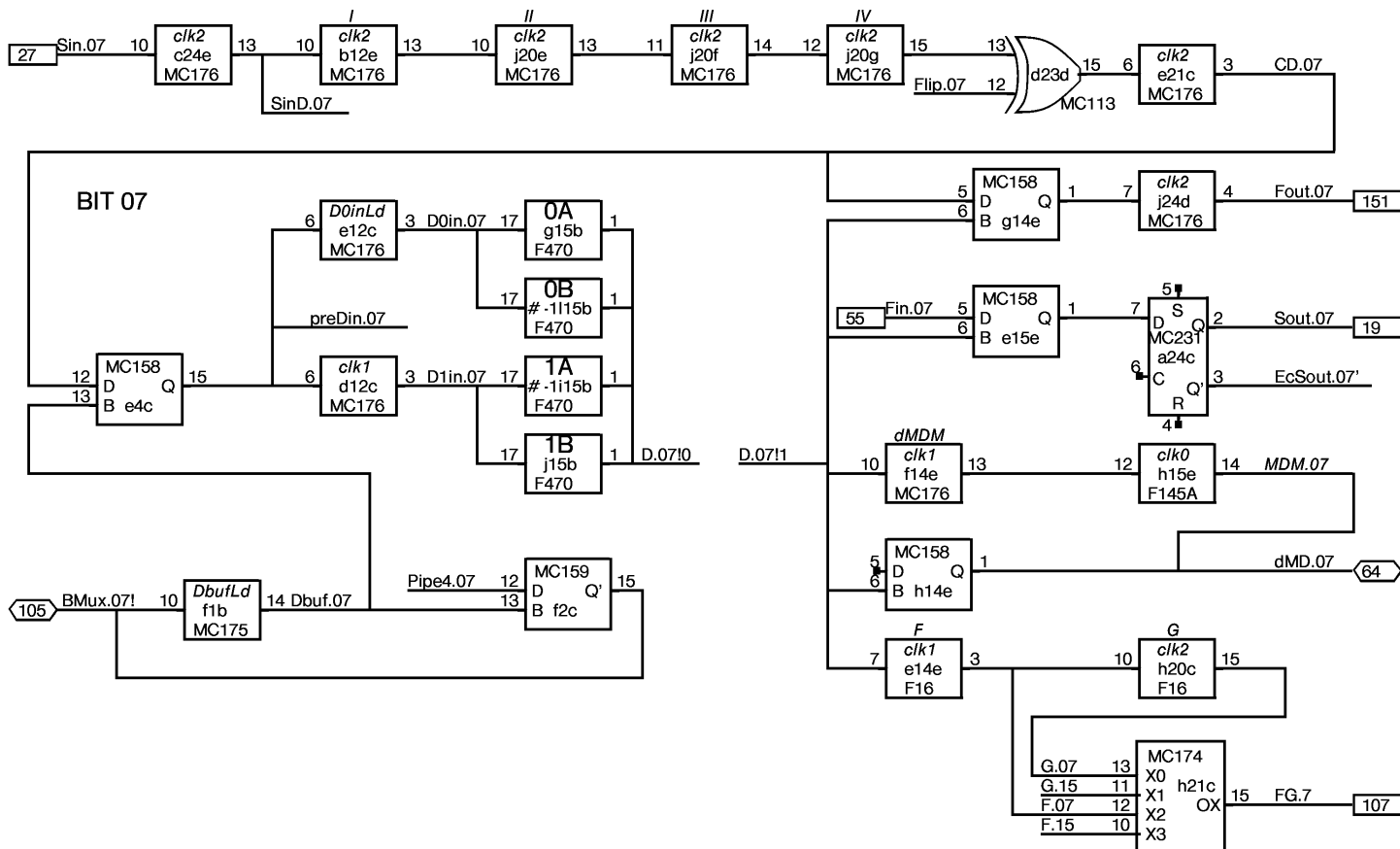
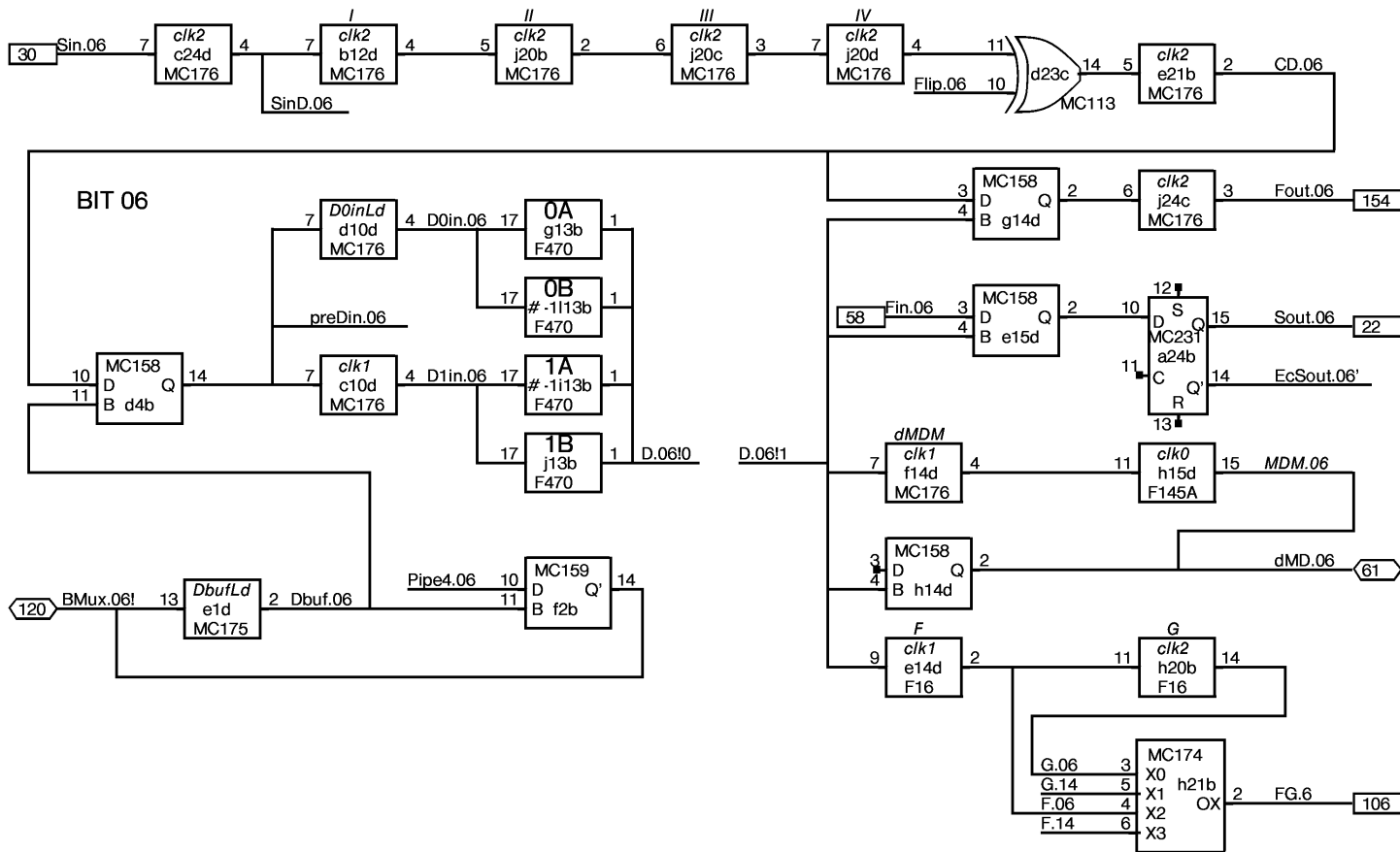
A single-bit error in Ci cannot yield two 1's in C456. C7 is computed so that parity of C0-7 is odd for any data with exactly one 1. Hence parity of Syn0-7 will be odd for any single bit error.

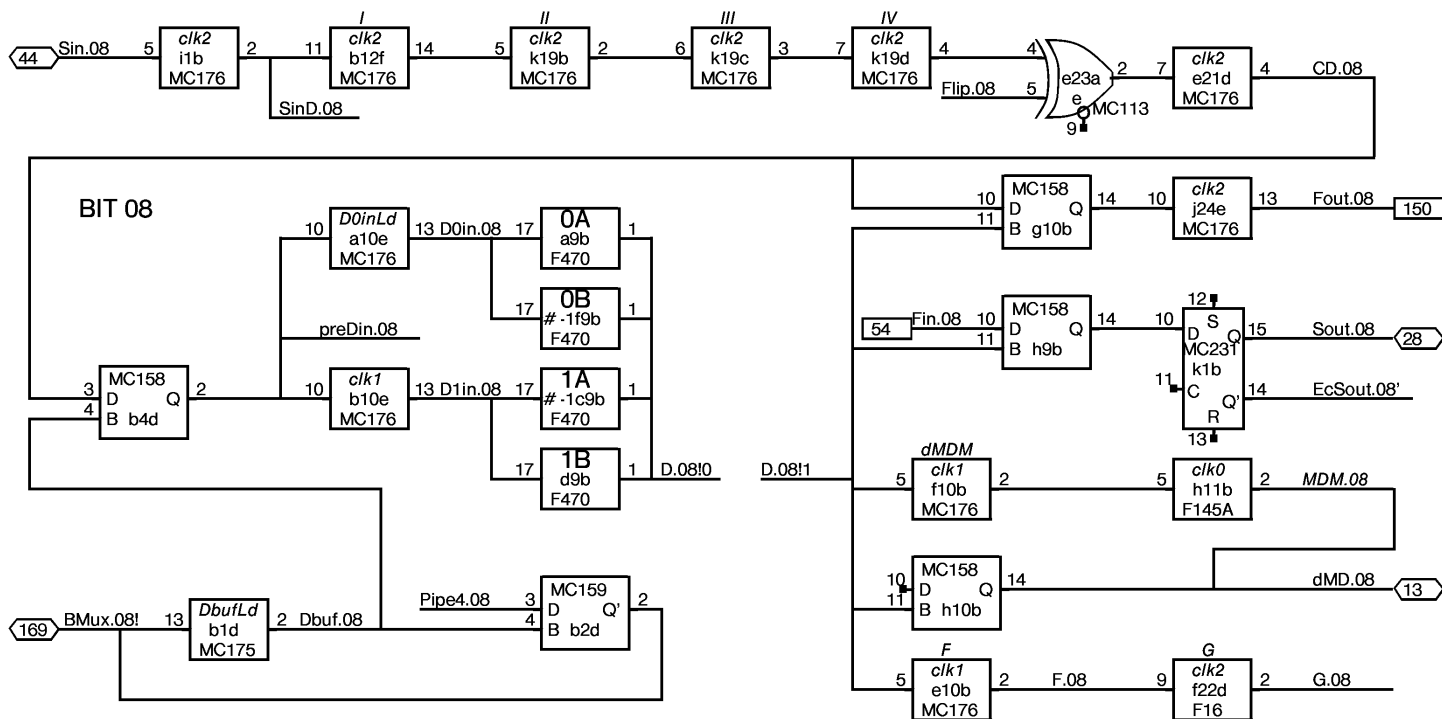


Bit 00 is on page 2

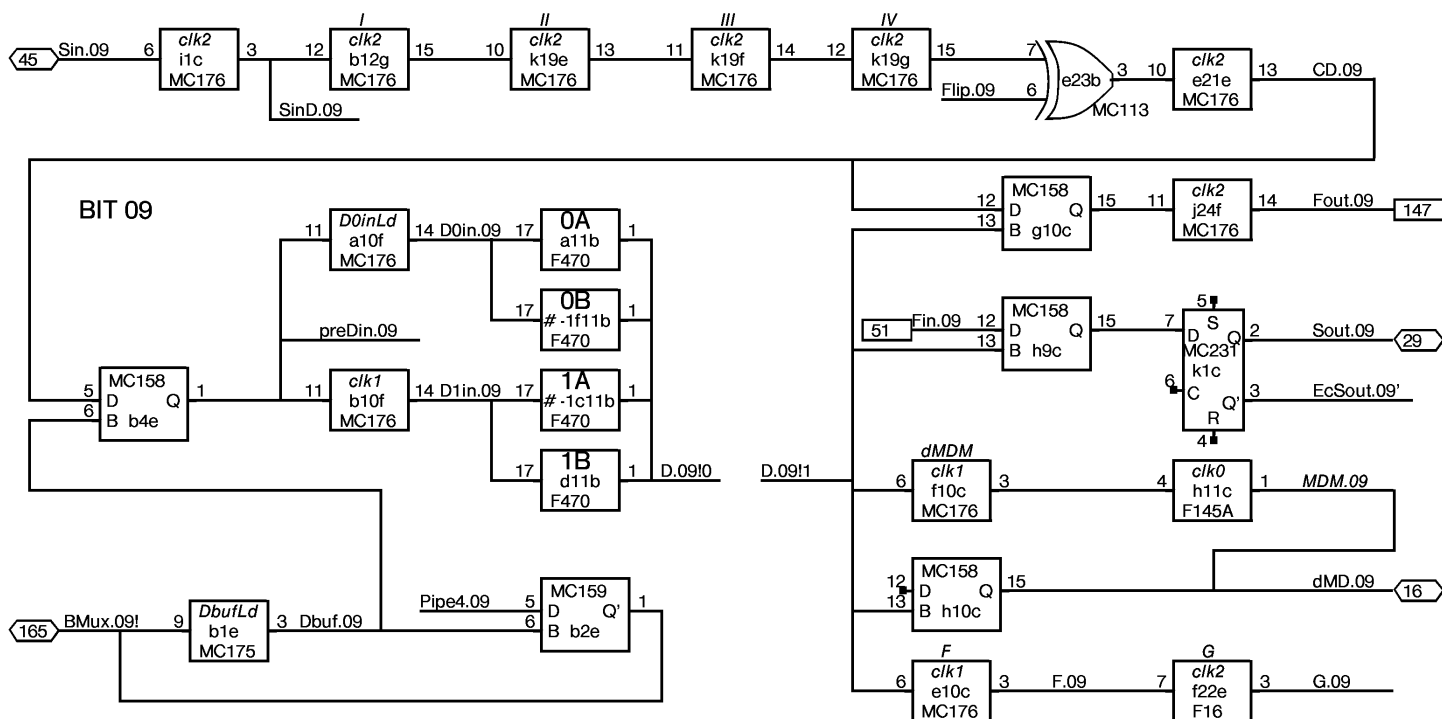


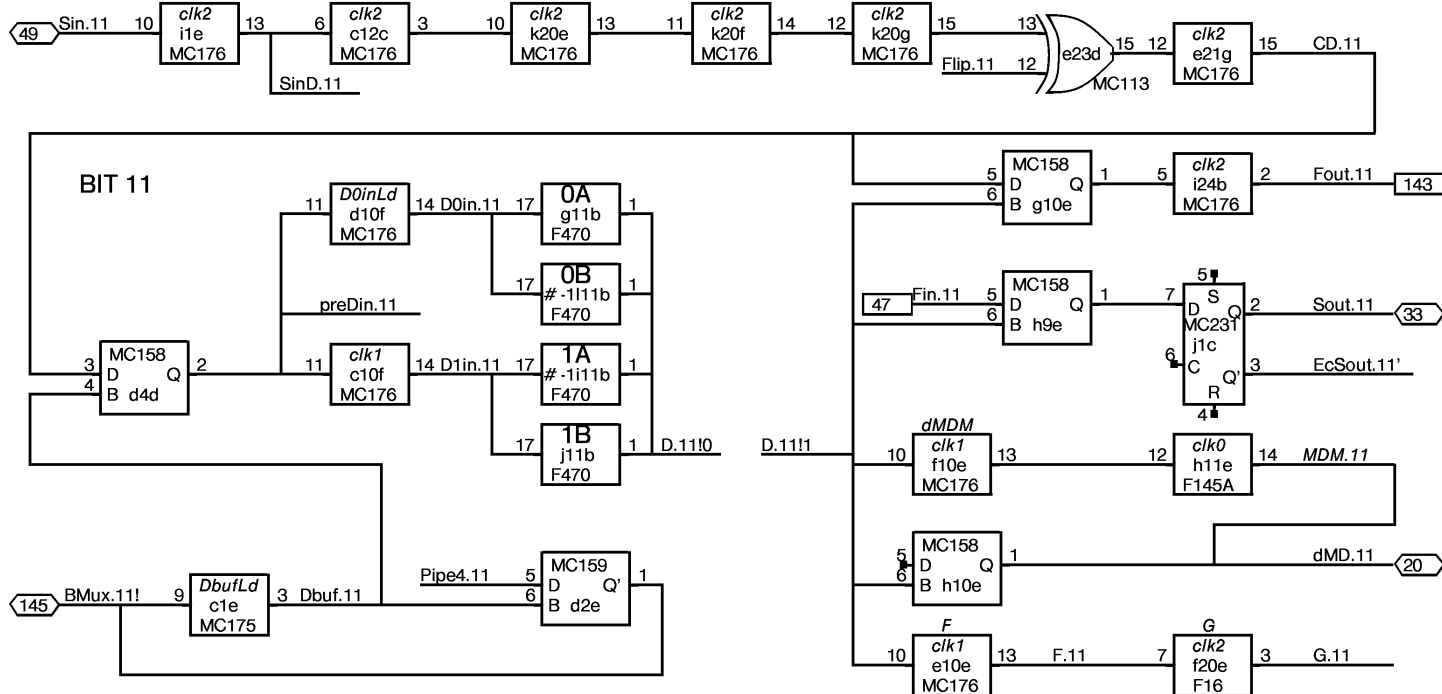
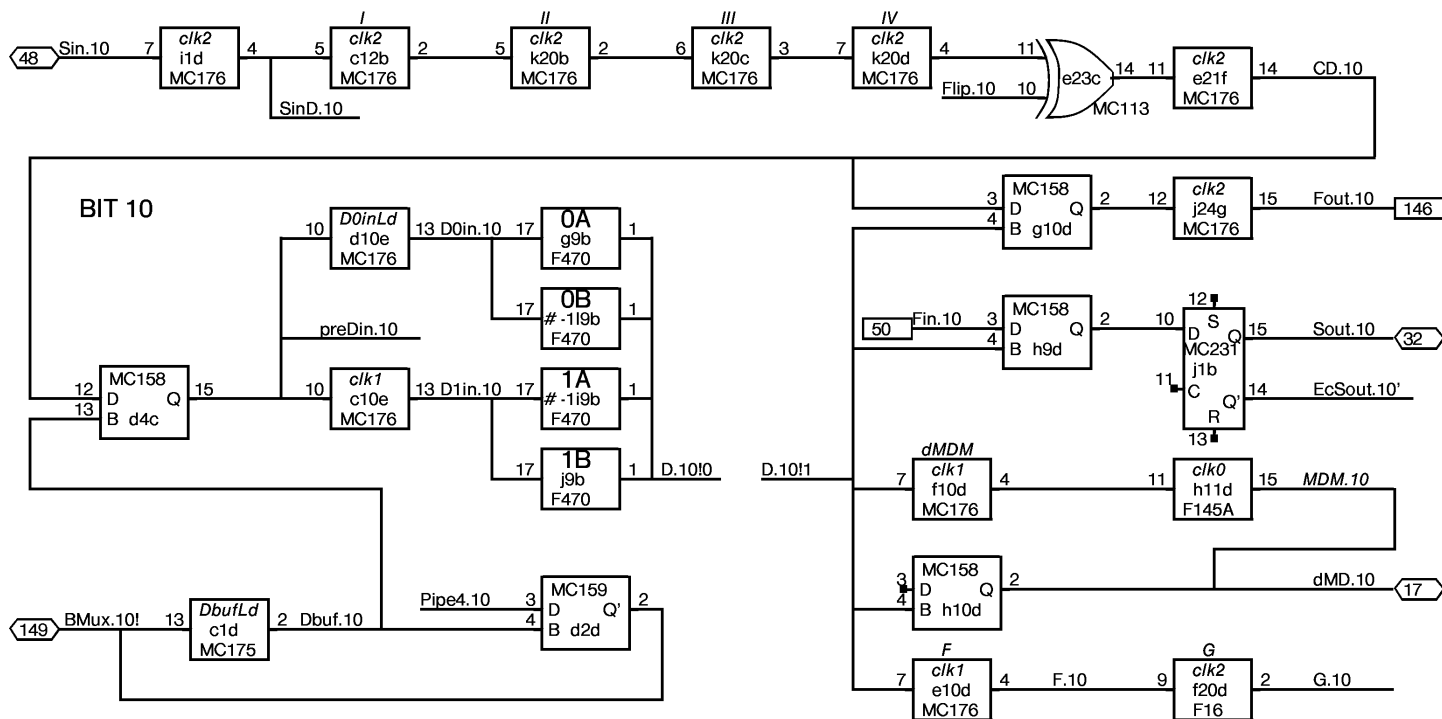


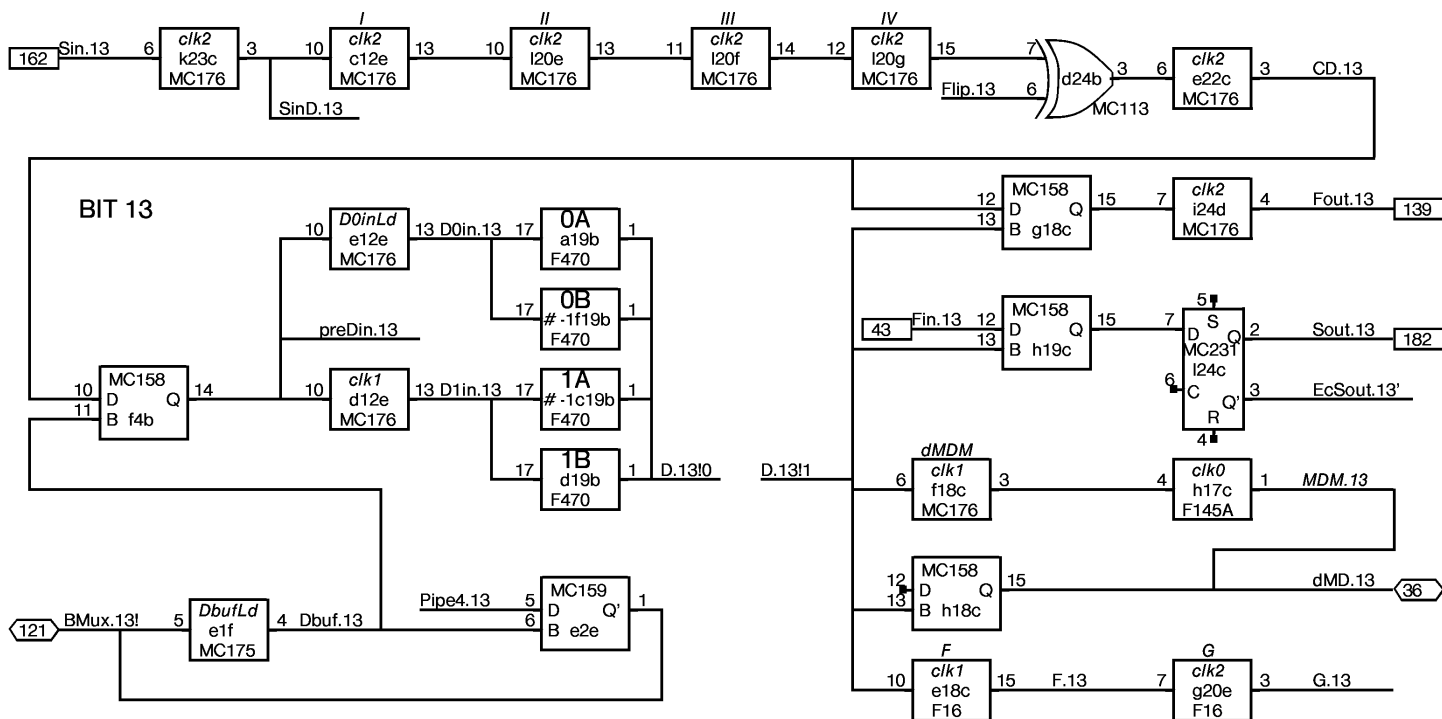
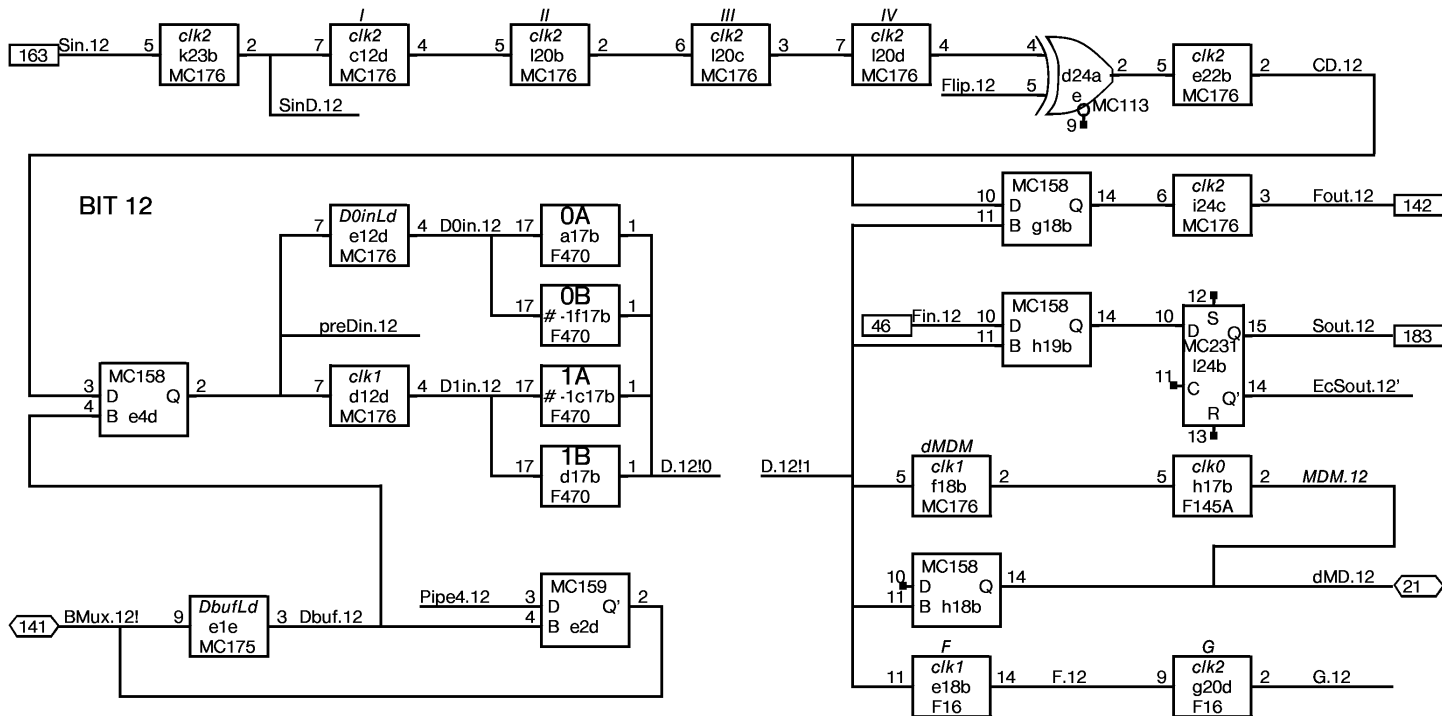


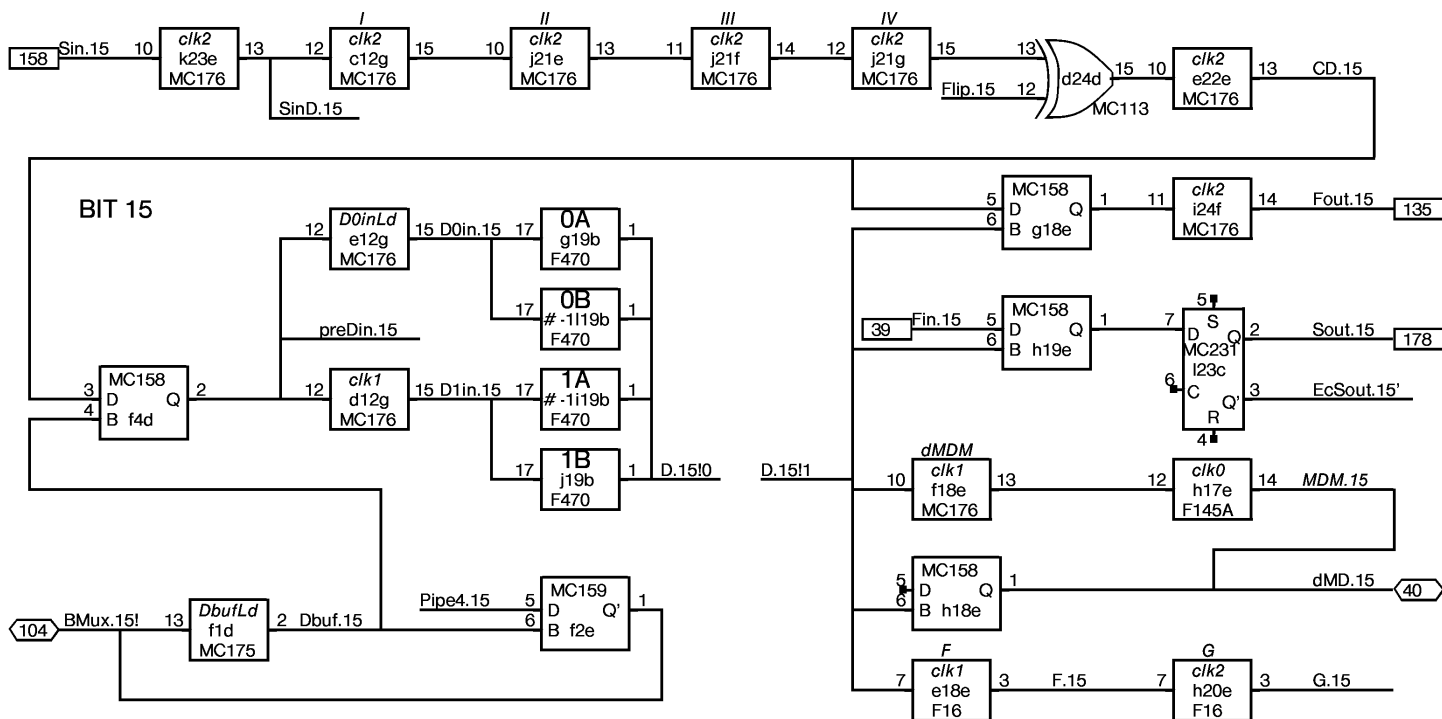
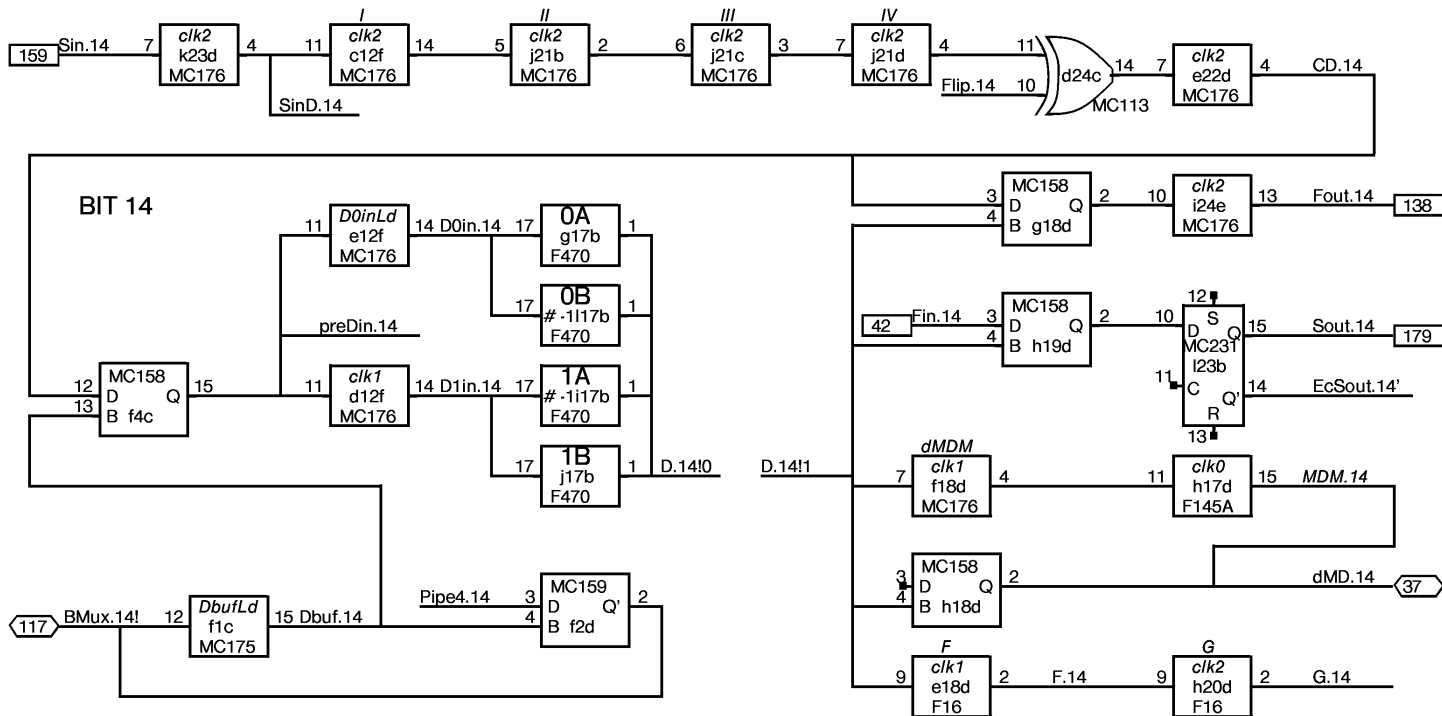


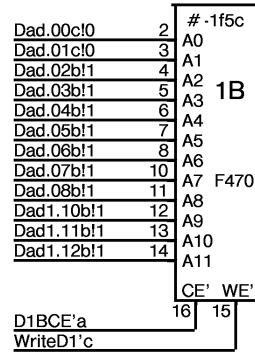
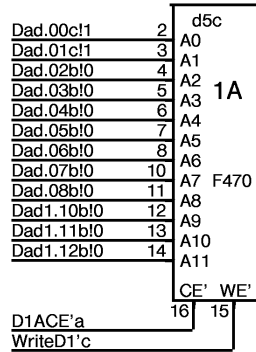
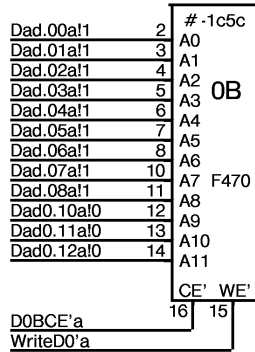
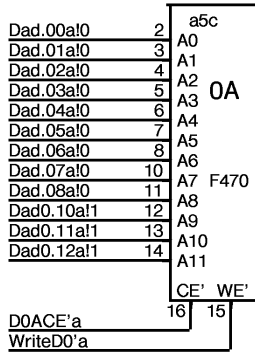
SURPRISE!
FG.8 is on bit slice 16



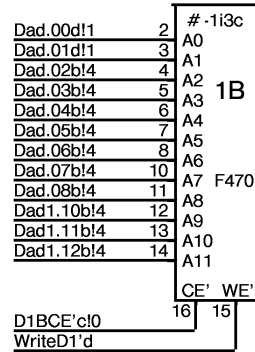
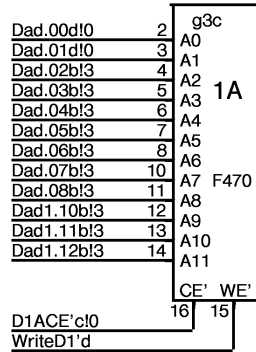
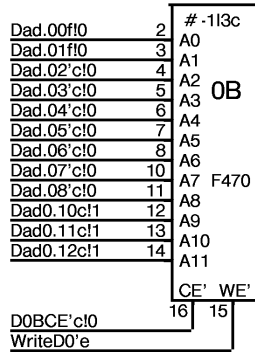
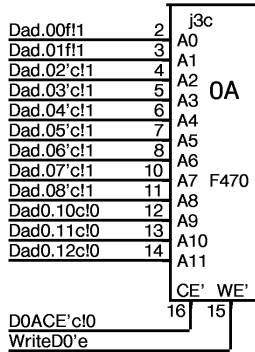




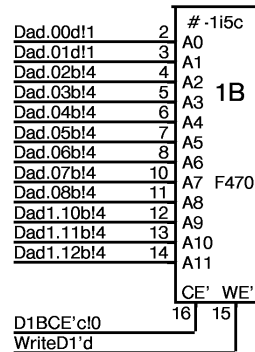
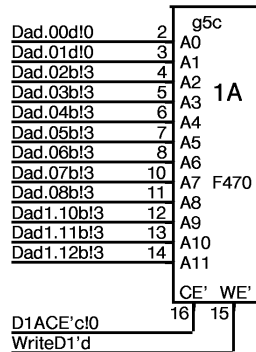
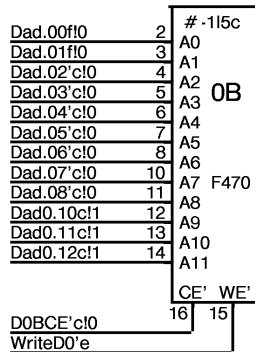
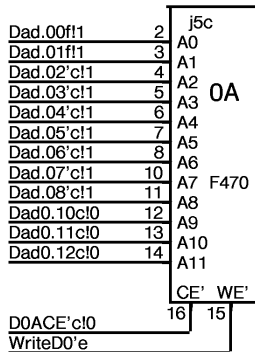




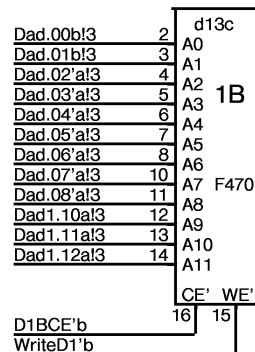
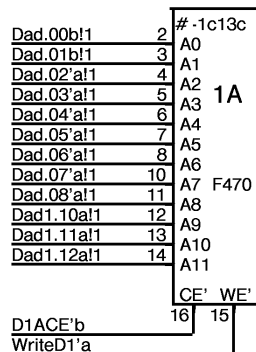
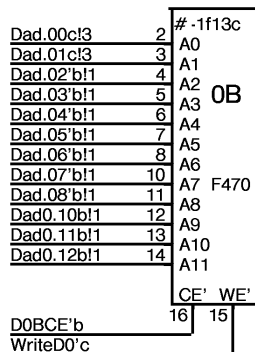
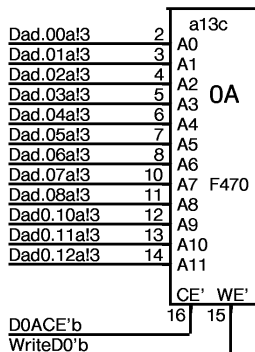
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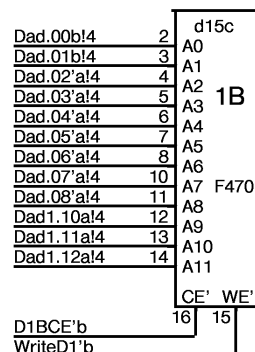
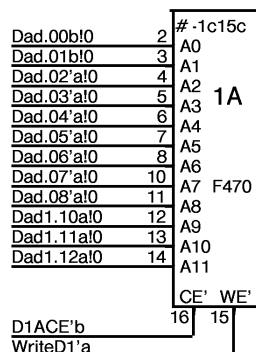
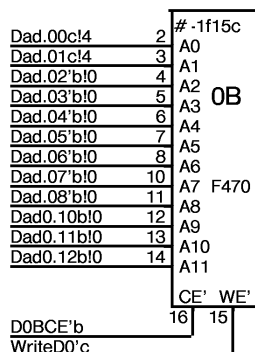
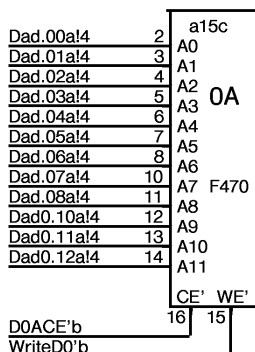
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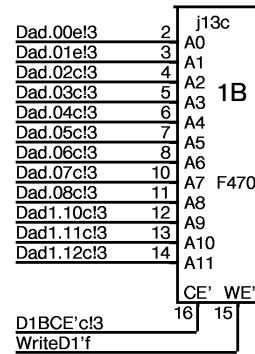
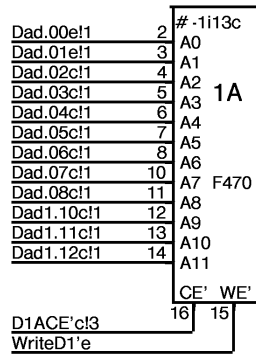
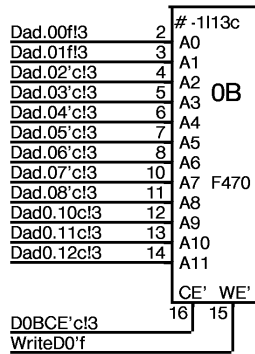
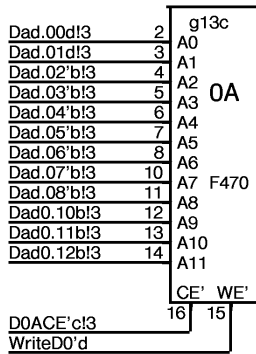
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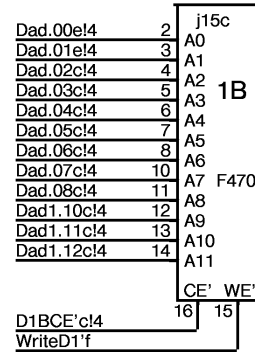
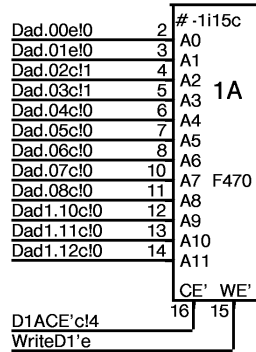
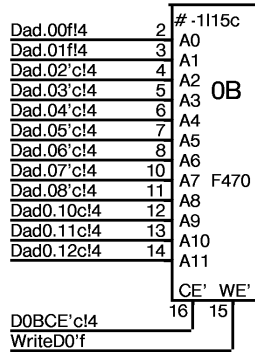
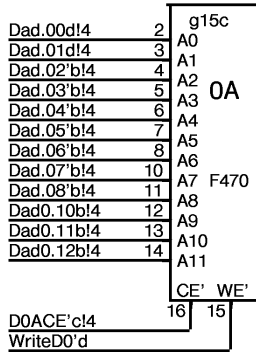
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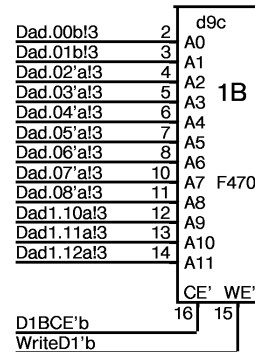
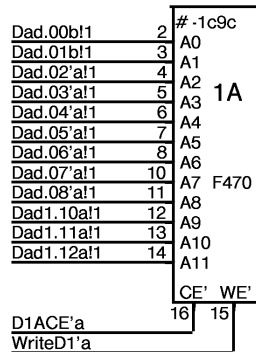
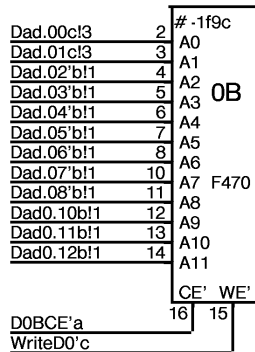
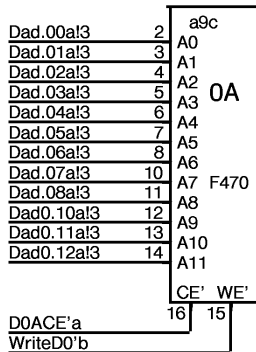
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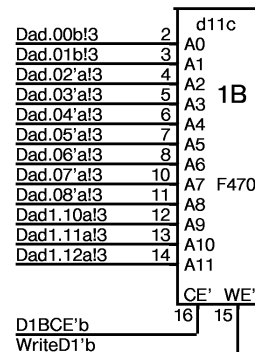
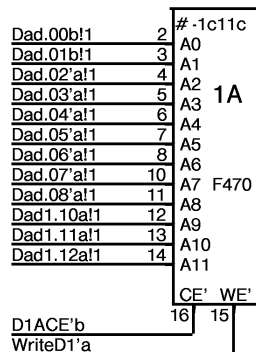
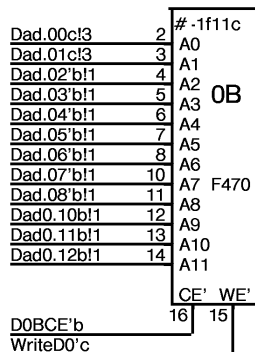
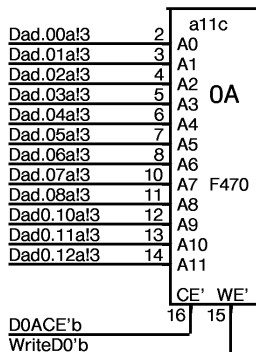
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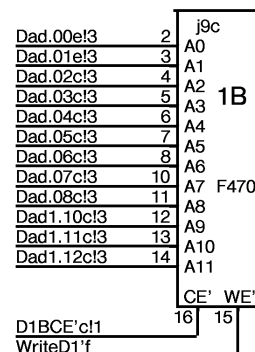
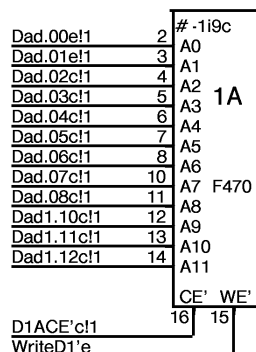
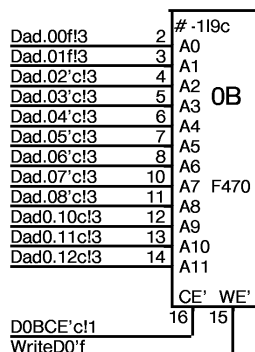
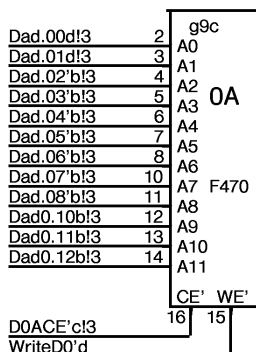
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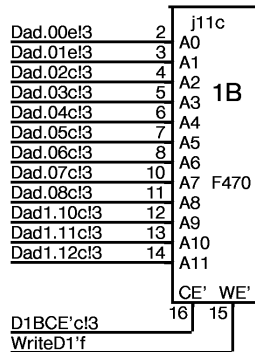
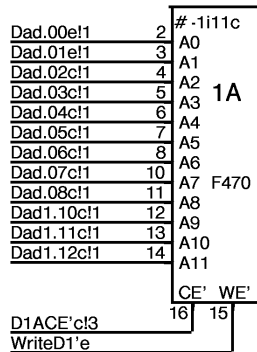
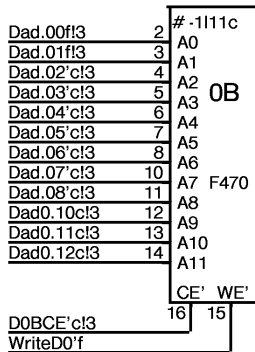
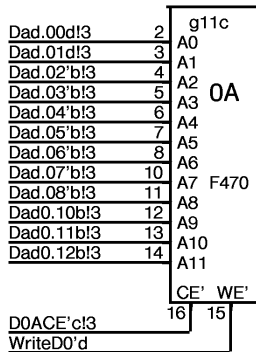
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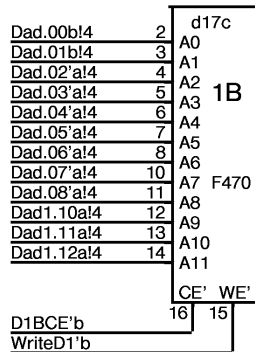
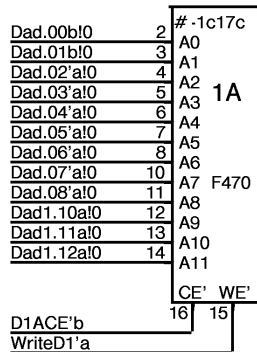
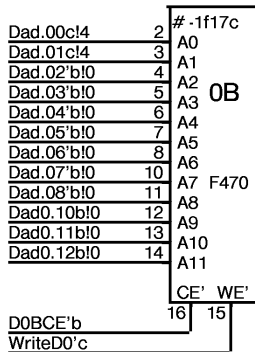
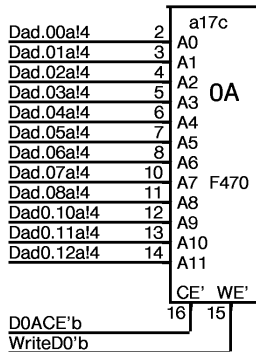
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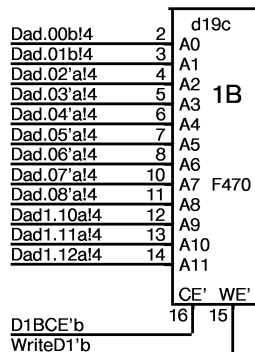
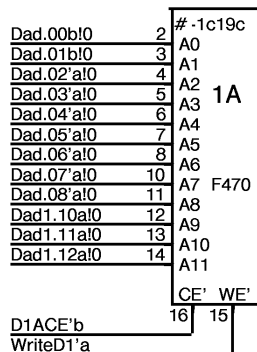
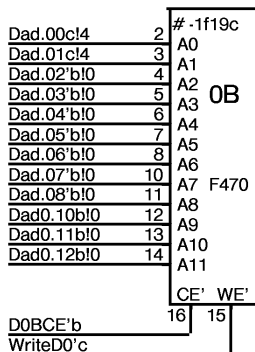
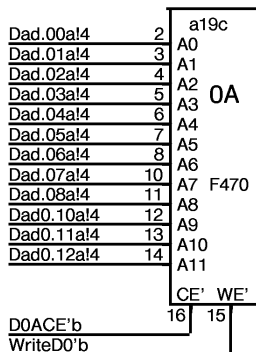
BIT 10



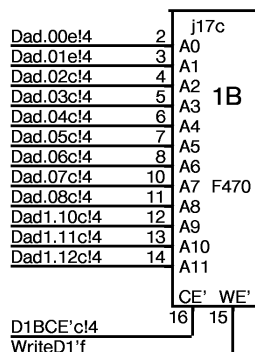
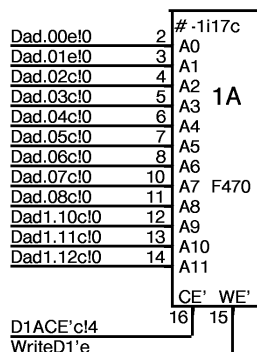
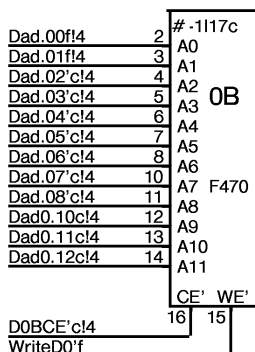
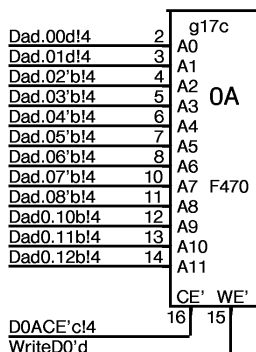
BIT 11



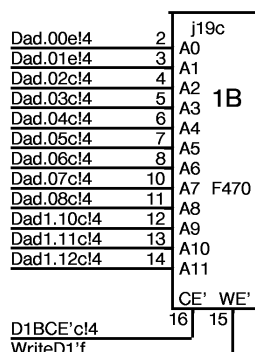
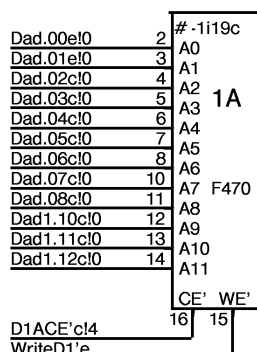
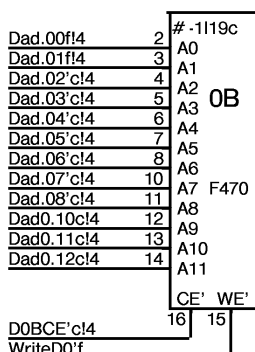
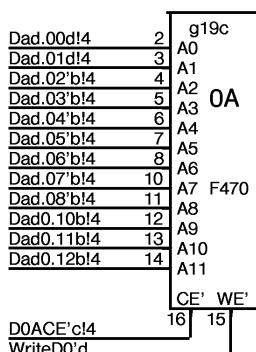
BIT 12



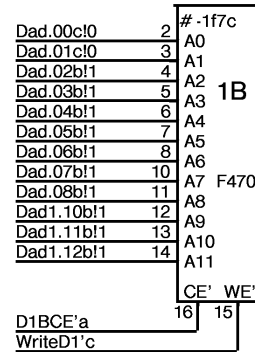
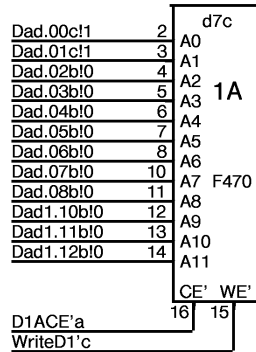
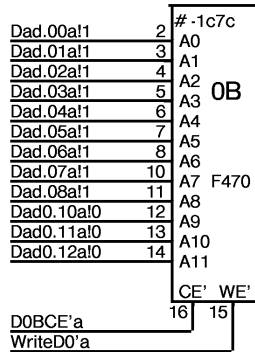
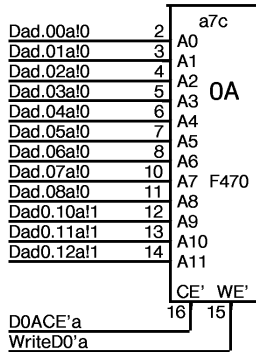
BIT 13



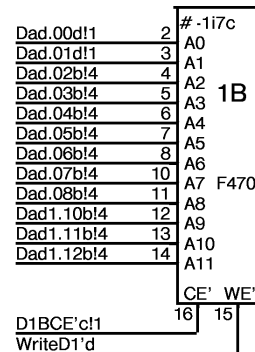
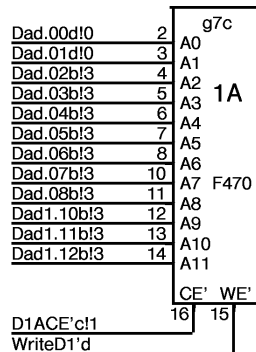
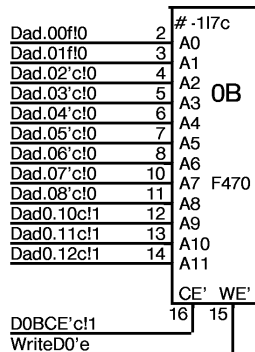
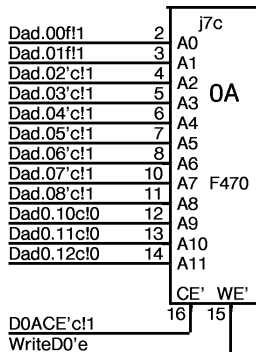
BIT 14



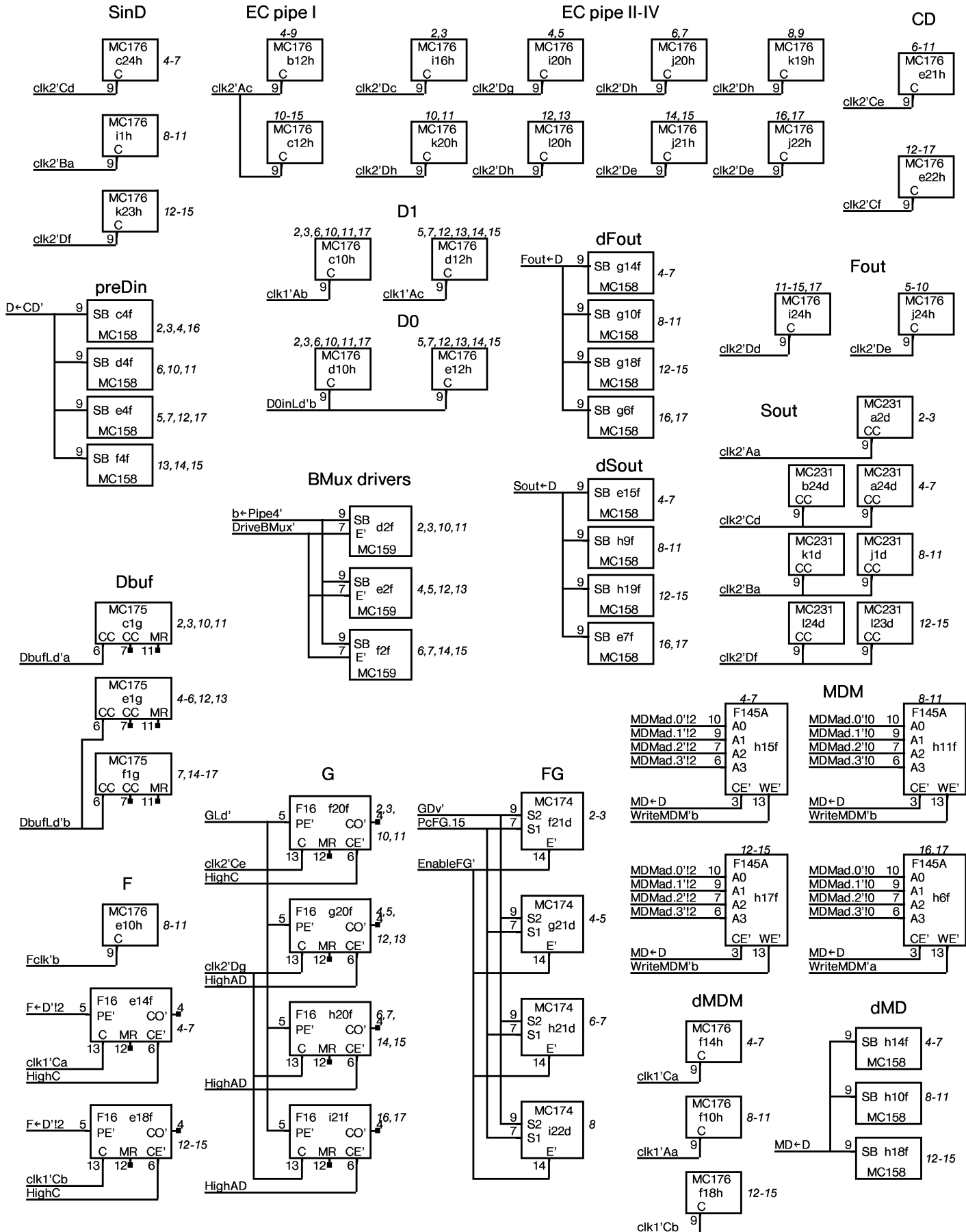
BIT 15

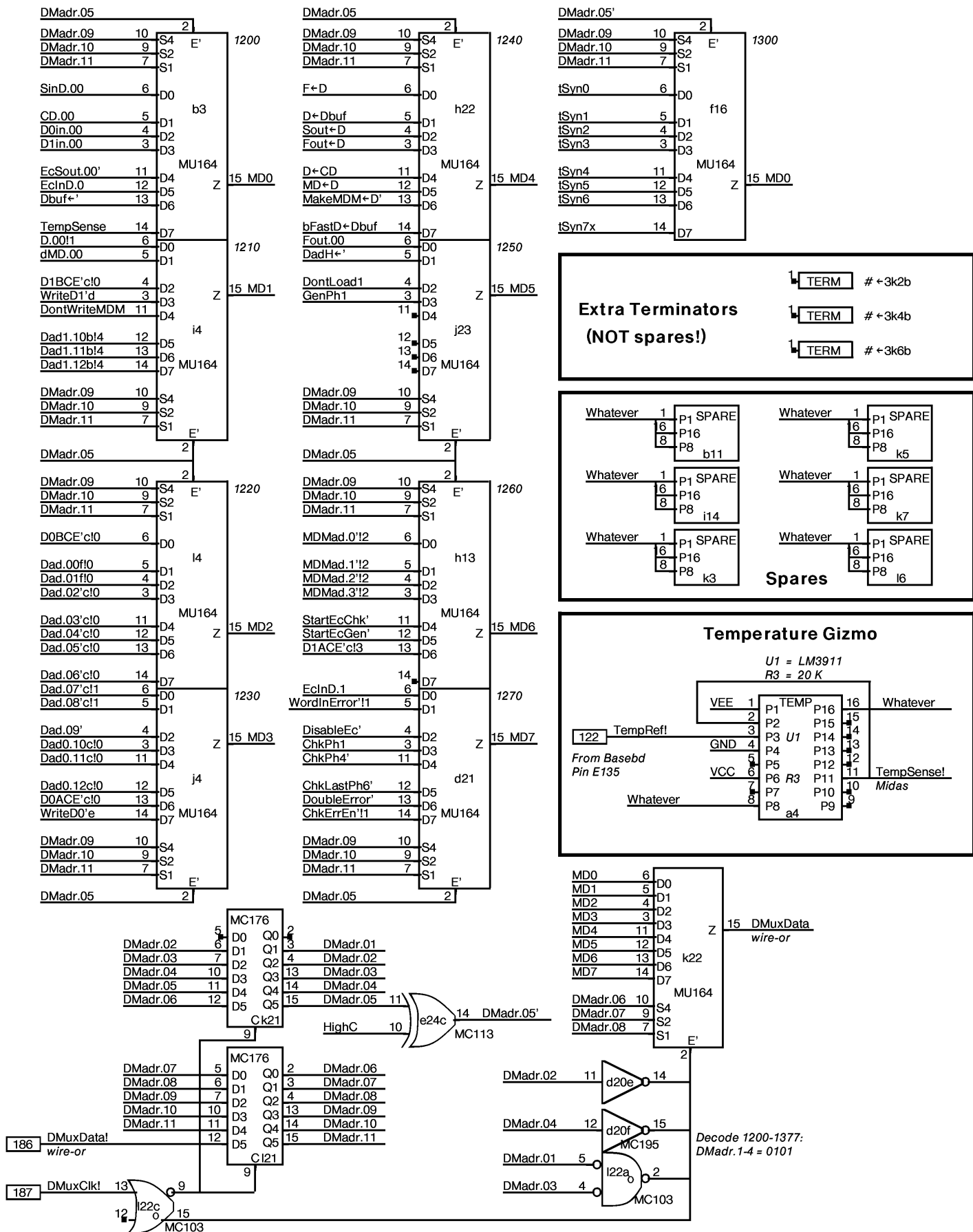


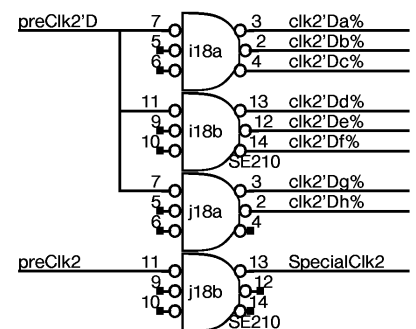
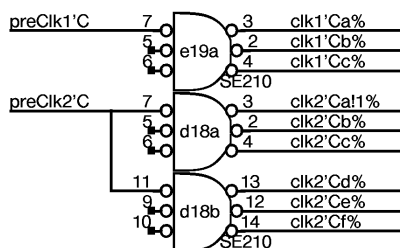
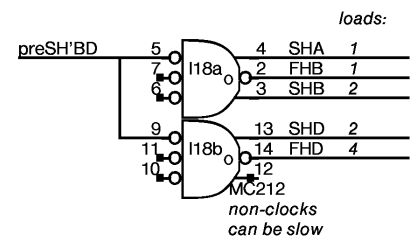
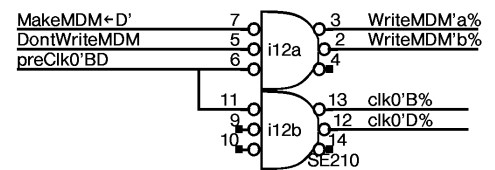
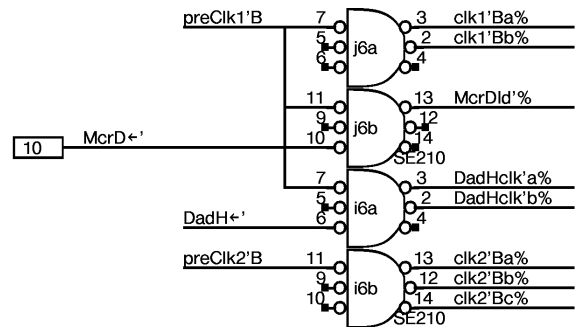
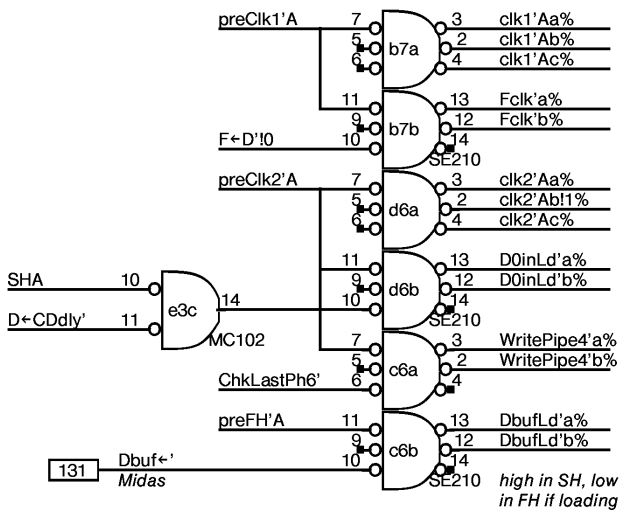
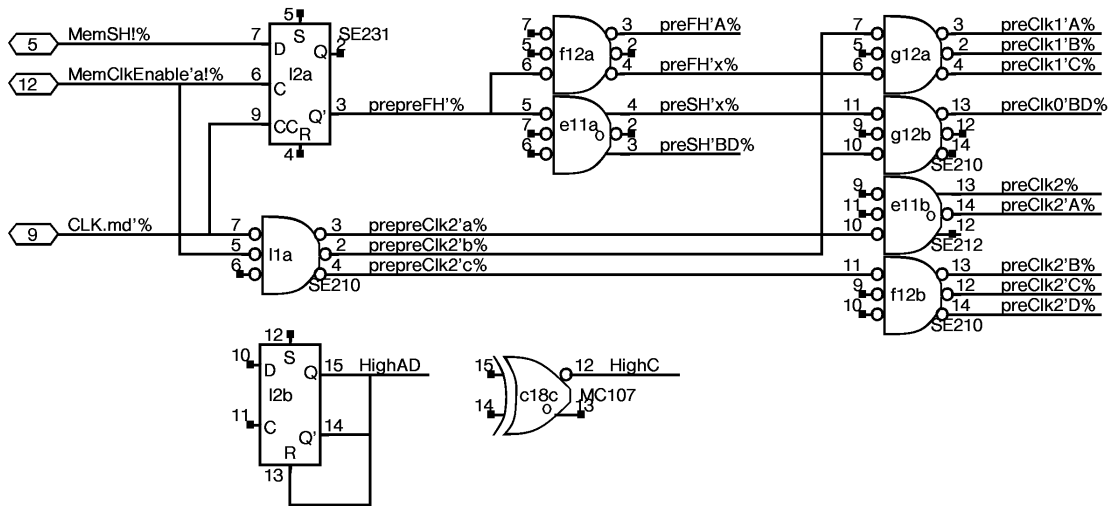
BIT 16



BIT 17







Cxxx														B	
A	a 181	b 168	c 153	d 137	e 124	f 109	g 93	h 80	i 64	j 48	k 33	l 20			
1	Sout drv 0-1 231 2a	BMux rcvr 0,1,8,9 175 *a	BMux rcvr 2,3,10,11 175 *a	EcOut 0,2,4,6 dSTPerr 176 2a	BMux rcvr 4-6,12,13 175 *b	BMux rcvr 7,14-17 175 *b	Dad rcvrs 2,3 231 *a	6,7 231 *b	SinD.8-11 EcSt.16,17 176 2a	Sout drv 10-11 231 2a	Sout drv 8-9 231 2a	Clock 210		1	
2	Sout drv 2-3 231 2a	BMux drv 0,1,8,9 159	SinD.0-3 EcInD.0 176 2a	BMux drv 2,3,10,11 159	BMux drv 4,5,12,13 159	BMux drv 6,7,14,15 159	4,5 231 *a	8,9 231 *b	0-5 TestSyn 176 *	Syn 6-8 176 *	TERM	231		2	
3	DOA.00 a,a,a,a,a 415A	SinD.0 CD.0 EcInD.0 Dbuf 10-11 159	DOB.00 a,a,a,a,a 415A	D1A.00 c,b,b,a,c 415A	MemError ECFault predInLd DriveBMux 102	D1B.00 c,b,b,a,c 415A	D1A.02 d,b,b,c,d 415A	Pipe34Ad DntWrtMDM 176 0	D1B.02 d,b,b,c,d 415A	DOA.02 f,'c,c,c,e 415A		DOB.02 f,'c,c,c,e 415A		3	
4	TempSense Gizmo PLAT	0,1,8,9 158	2,3,4,16 158	predIn 6,10,11 158	5,7,12,17 158	13,14,15 158	dFout 0-3 158	MdM 0-3 145 *a	CE1B D.0 10-12(1) Write1 DWrMDM dMD.0	Dad.7-9 CE0A, 10-12(0) Write0	TERM	Dad.0-6 CE0B MU164		4	
5	DOA.01 a,a,a,a,a 415A	12-15 145	DOB.01 a,a,a,a,a 415A	D1A.01 c,b,b,a,c 415A	dSout 0-3 158	D1B.01 c,b,b,a,c 415A	D1A.03 d,b,b,c,d 415A	dMd 0-3,16,17 197	D1B.03 d,b,b,c,d 415A	DOA.03 f,'c,c,c,e 415A		DOB.03 f,'c,c,c,e 415A		5	
6	4-7 Pipe4 145	8-11 145	clks 210	clks 210	F 0-3,16,17 176 *a	dMdM 0-3,16,17 176 1a	dFout 16,17 158	MdM 16,17 145 *a	clks 210	clks 210	TERM			6	
7	DOA.16 a,a,a,a,a 415A	clks 210	DOB.16 a,a,a,a,a 415A	D1A.16 c,b,b,a,c 415A	dSout 16,17 158	D1B.16 c,b,b,a,c 415A	D1A.17 d,b,b,c,d 415A	Dad1 10-12 b F16 1a	D1B.17 d,b,b,c,d 415A	DOA.17 f,'c,c,c,e 415A		DOB.17 f,'c,c,c,e 415A		7	
8	Dad0 10-12 a F16 2b0	Dad.2-5 a,'a 101	Dad.6-8 a,'a 101	Dad.0-1 a,b,c 210	Writes a,b,c 211	Dad.2-5 b,'b 101	Dad.6-8 b,'b 101	Writes d,e,f 210	Dad.0-1 d,e,f 210	Dad.2-5 c,'c 101	Dad.6-8 c,'c 101	Dad0 13 10-12 c F16 2b		8	
9	DOA.08 a,a,a,a,b 415A	Dad1 10-12 a F16 1b	D1A.08 b,'a,a,a,a 415A	D1B.08 b,'a,a,b,b 415A	Dad0 10-12 b F16 2b2	DOB.08 c,'b,b,a,c 415A	DOA.10 d,'b,b,c,d 415A	dSout 8-11 158	D1A.10 e,c,c,c,e 415A	D1B.10 e,c,c,c,f 415A	Dad1 10-12 c F16 1b	DOB.10 f,'c,c,c,f 415A		9	
10	DOin 0,1,4, 8,9,16 176 *a	D1in 0,1,4, 8,9,16 176 1b	D1in 2,3,6, 10,11,17 176 1b	DOin 2,3,6, 10,11,17 176 *b	F 8-11 176 *b	dMdM 8-11 ClearWA 176 1a	dFout 8-11 158	dMd 8-11 158	CE's 0A, 1A 210	CE's 0B, 1B 210	preCE 159	DntLoad0 1664		10	
11	DOA.09 a,a,a,b,b 415A		D1A.09 b,'a,a,b,a 415A	D1B.09 b,'a,a,b,b 415A	preclks 212	DOB.09 c,'b,b,b,c 415A	DOA.11 d,'b,b,c,d 415A	MdM 8-11 145 *b	D1A.11 e,c,c,c,e 415A	D1B.11 e,c,c,c,f 415A	B 180	DOB.11 f,'c,c,c,f 415A		11	
12	0-3,17 176 2c	Ec pipe I 4-9 176 2c	10-15 176 2c	D1in 5,7,12 13,14,15 176 1c	DOin 5,7,12 13,14,15 176 *b	preclks 210	preclks 210	MDMad 141 2c	WriteMDM clk0'B,D 210		Generator A 180	163		12	
13	DOA.04 a,a,a,b,b 415A	Cphi 172	D1A.04 b,'a,a,b,a 415A	D1B.04 b,'a,a,b,b 415A	ChkPhi F00 2a0	DOB.04 c,'b,b,b,c 415A	DOA.06 d,'b,b,c,d 415A	MDMad(4) CE1A StrEcChk StrEcGd 415A	D1A.06 e,c,c,c,e 415A	D1B.06 e,c,c,c,f 415A		DOB.06 f,'c,c,c,f 415A		13	
14	0-7 163	8-15 163	SinD.17 170	ChkQWi ChkWi F16 2a0	F 4-7 F16 1a	dMDM.4-7 LastQWi 176 1a	dFout 4-7 158	dMd 4-7 158		C 180	D 180	E 180		14	
15	DOA.05 a,a,a,b,b 415A	A 180	D1A.05 b,'a,a,b,a 415A	D1B.05 b,'a,a,b,b 415A	dSout 4-7 158	DOB.05 c,'b,b,b,c 415A	DOA.07 d,'b,b,c,d 415A	MdM 4-7 145 *b	D1A.07 e,c,c,c,e 415A	D1B.07 e,c,c,c,f 415A	F 180	DOB.07 f,'c,c,c,f 415A		15	
16	B 180	Checker C 180	D 180	A-C 158	D-E 158	tSyni MU164	Pipell-IV 0,1 176 2c	Chk/Gen 3 dClearWA 104	Pipell-IV 2,3 176 2c	(D) D+CDdly 231 2b	A-C 158	(E) 231 2a		16	
17	DOA.12 a,a,a,b,b 415A	E 180	D1A.12 b,'a,a,b,a 415A	D1B.12 b,'a,a,b,b 415A	ChkPh,etc. SinD.16dly 176 2a2	DOB.12 c,'b,b,b,c 415A	DOA.14 d,'b,b,c,d 415A	MdM 12-15 145 *b	D1A.14 e,c,c,c,e 415A	D1B.14 e,c,c,c,f 415A	D-E 158	DOB.14 f,'c,c,c,f 415A		17	
18	tSyn0-3 F00 2b	tSyn4,7 231 2b	SinD.16 D+CD HighC 107	clks 210	F 12-15 F16 1b	dMdM 12-15 TickTockCE ChkLstPh6 176 1b	dFout 12-15 158	dMd 12-15 158	clks 210	clks 210	GenPhi F00 2a	FH, SH 212		18	
19	DOA.13 a,a,a,b,b 415A	tSyn5,6 231 2b	D1A.13 b,'a,a,b,a 415A	D1B.13 b,'a,a,b,b 415A	clks 210	DOB.13 c,'b,b,b,c 415A	DOA.15 d,'b,b,c,d 415A	dSout 12-15 158	D1A.15 e,c,c,c,e 415A	D1B.15 e,c,c,c,f 415A	Pipell-IV 8,9 176 2h	DOB.15 f,'c,c,c,f 415A		19	
20	Syni F16 2c	Corrector WordnError F00 2c	Syn3 D+Dbufdly 231 2c	EcCheck stuff MU164	CD 0-5 176 2e	G 2,3,10,11 F16 2e	G 4,5,12,13 F16 2g	G 6,7,14,15 F16 2g	Pipell-IV 4,5 176 2g	Pipell-IV 6,7 176 2h	Pipell-IV 10,11 176 2h	Pipell-IV 12,13 176 2h		20	
21	DbIError 170	Error ChkErrEn' 109	Syn3 D+Dbufdly 231 2c	EcCheck stuff MU164	6-11 176 2e	FG 2,3 174	FG 4,5 174	FG 6,7 174	G 16,17 F16 2g	Pipell-IV 14,15 176 2e	1-5 Midas 176	6-11 176		21	
22	Flips16,17 ChkErrEn' 102	Flips 0-7 162	Flips 8-15 162	dCD 0-3 113	12-17 176 2f	G 0,1,8,9 F16 2f	preWrite 231 *	X+Ys MU164	FG 8 174	Pipell-IV 16,17 176 2e	DMuxData 164	Midas 2 Bank0CE 103		22	
23	4-7 F16 1c	dPipe4 8-11 F16 1c	12-15 F16 1c	4-7 113	8-11 113	FG 0,1 174	dMD+D DntLd1 dD+Dly 106	preWrite bTransport 102	DadH+ 231 0	Fout.0 DntLd1 DadH+ GenPh1	SinD.12-15 D+CDdly 176 2f	Sout drv 14-15 231 2f		23	
24	Sout drv 6-7 231 2d	Sout drv 4-5 231 2d	SinD.4-7 EcInD.1 ChkPh5w.o. 176 2d	12-15 113	16,17 DMadr.05' 113	EcOut 1,3,5,7 D+dlly' 176 2f	MakeX rcvr 176 2d	bFastD+Dbf preWrite 1664	11-15,17 176 2d	Fout drivers 5-10 176 2e	0-4,16 176 2f	Sout drv 12-13 231 2f		24	

Exxx														D	
C	a 11	b 26	c 39	d 55	e 70	f 86	g 99	h 114	i 129	j 143	k 159	l 174			

D chip code: Dad.0-1,2-8,10-12,CE,Write

XEROX PARC/CSL	Project Dorado	Reference Memory data board Layout	File MemD23.sil	Designer D. Clark	Rev Ca	Date 7/11/79	Page 23
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